Production Beyond the Firm: The Aggregate Value of Subcontracting

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Why do firms subcontract production?

- Firms decide how much to produce and which parts of **production to handle themselves**
  - **Subcontracting**: paying another firm to perform some production tasks
  - Example: a garment firm sends cut fabric to a workshop that stitches the garments

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- **Specialization**: other firms perform some tasks better
  - The workshop stitches with specialized equipment and methods
- **Flexibility**: firms change production size without adjusting in-house inputs
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$\Rightarrow$ Trade-off: adjusting size through subcontracting saves adjustment costs, but lowers efficiency
  - When demand rises sharply, the garment firm also delegates the cutting, which the workshop does worse

Research questions

How much does subcontracting contribute to **aggregate productivity**, and through **which motive**?

How does accounting for subcontracting change the **measured consequences** of input **adjustment frictions**?

This paper

- Empirics: Portuguese firm-level administrative data
  - Subcontracting: **12%** of total costs, used by 45% of firms
  - Top vs bottom sales decile: **70%** vs **21%** of firms subcontract, 20% vs 9% of their costs
  - Subcontracting tracks sales the closest: **1.4**, vs 1 for materials and below 0.4 for labor and capital

This paper

- Empirics: Portuguese firm-level administrative data
  - ⇒ Subcontracting: a sizeable and unique input, used to adjust production size
- Model: firm dynamics with **costly labor adjustment** and **task delegation**
  - Firms differ in how much of their production they can delegate
  - **Specialization**: a subcontractor performs some tasks better
  - **Flexibility**: delegated tasks need no in-house adjustment costs
  - The subcontractor faces the same frictions, but pools its clients' shocks
    - ⇒ Flexible production with worker protection

This paper

- Empirics: Portuguese firm-level administrative data
  - $\Rightarrow$ Subcontracting: a sizeable and unique input, used to adjust production size
- Model: firm dynamics with **costly labor adjustment** and **task delegation**
  - $\Rightarrow$ Two reasons to subcontract: specialization and flexibility
- Quantification: calibrated model
  - Subcontracting raises aggregate productivity by **3.9%**
    - Specialization: **81%** of the gain; flexibility: **19%**
  - Subcontracting lets firms absorb idiosyncratic shocks, but hardly aggregate ones
  - Ignoring subcontracting overstates the losses from adjustment frictions
    - Productivity losses of **7.1%** instead of **3.5%**
  - Subcontracting also determines who the largest firms are

Contribution

- Costs of adjustment frictions

  Hopenhayn and Rogerson (1993); Cooper and Haltiwanger (2006); Asker et al. (2014); Roys (2016); Boar et al. (2025)

  ⇒ Evaluating the costs of frictions requires the **margins firms use to avoid them**

- Intermediates and primary inputs

  Huneeus et al. (2021); Dhyne et al. (2024); Hubmer et al. (2025); Mertens and Schoefer (2026)

  ⇒ Shift from labor to intermediates is largely **subcontracting**

  ⇒ Subcontracting as a margin to **adjust production size**

- Outsourcing and firm boundaries

  Grossman and Helpman (2002); Antràs (2003); Antràs and Helpman (2004); Boehm and Oberfield (2020); Rubbo (2023); Baqaee et al. (2025); Goldschmidt and Schmieder (2017); Bilal and Lhuillier (2021); Arnarson et al. (2026)

  ⇒ **Specialization** and **flexibility** gains, measured separately

- Flexible labor arrangements

  Bentolila and Saint-Paul (1992); Cahuc and Postel-Vinay (2002); Caggese and Cuñat (2008); Cahuc et al. (2016); Autor (2003); De Leon et al. (2024); Pijoan-Mas and Roldan-Blanco (2025); Bostanci (2026)

  ⇒ Flexible margin, but, unlike fixed-term contracts, valued mostly for **specialization**

Empirics

Model

Calibration

Quantification

Data and variable definition

- **SCIE** (*Sistema de Contas Integradas das Empresas*)
  - Administrative balance sheet data at firm-year level
  - Universe of non-financial, non-insurance, and non-public sector firms operating in Portugal
  - Detailed input expenditure data from 2010 onwards

Data and variable definition

- **SCIE** (*Sistema de Contas Integradas das Empresas*)

  $\Rightarrow$ Administrative balance sheets of Portuguese firms, 2010–2022

- Intermediate inputs can be separated into seven categories:
  1. **Subcontracting**
  2. Materials
  3. Specialized services
  4. Short-lived tools and supplies
  5. Energy and fluids
  6. Transportation
  7. Other services

Data and variable definition

- **SCIE** (*Sistema de Contas Integradas das Empresas*)
  - ⇒ Administrative balance sheets of Portuguese firms, 2010–2022
- Intermediate inputs
  - ⇒ Subcontracting, materials, specialized services, short-lived tools and supplies, energy and fluids, transportation, other services
- What counts as subcontracting

	Task of the firm's production process	Provider's own workers, equipment and methods	Example: garment firm
Subcontracting	✓	✓	A workshop stitches the garments
Materials	✗	✗	The fabric
Staffing	✓	✗	Agency workers on its line
Support services	✗	✓	Cleaning, legal services

FACT 1. Subcontracting is widespread

Aggregate

12.4%

of total costs

Total costs: labor, capital and intermediates

All inputs | By sector | Intermediates by sector

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Total costs: labor, capital and intermediates

Extensive margin

44.9%

of firms subcontract

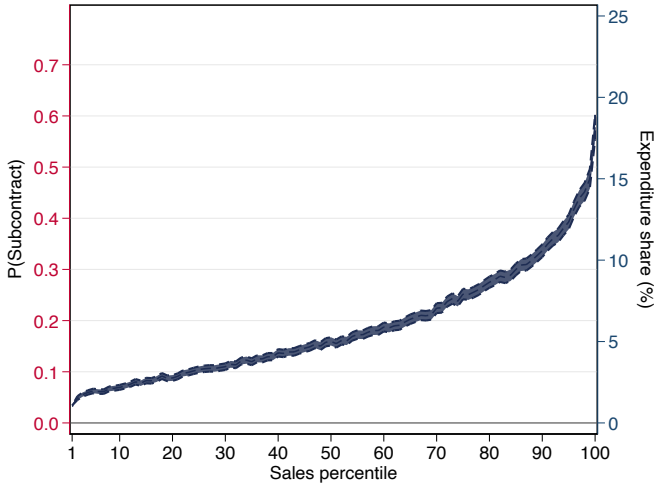
Intensive margin

17.6%

of their costs, among firms that subcontract

All inputs | By sector | Intermediates by sector

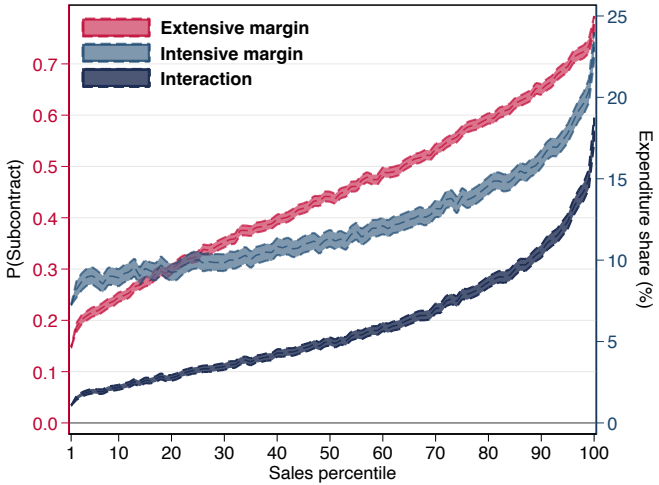
FACT 2. Larger firms subcontract more, on both margins



Notes: sales percentiles within sector × year, controlling for sector × year FEs.

Regression | Main inputs | Intermediates | Between vs within | Within firm | Firm age | By sector | Other countries

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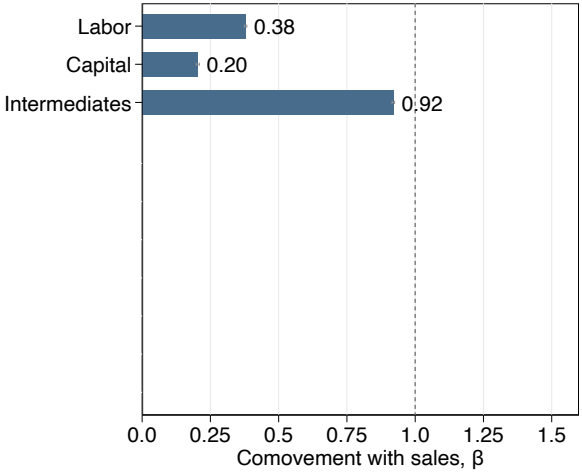


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FACT 3. Subcontracting moves the most with sales

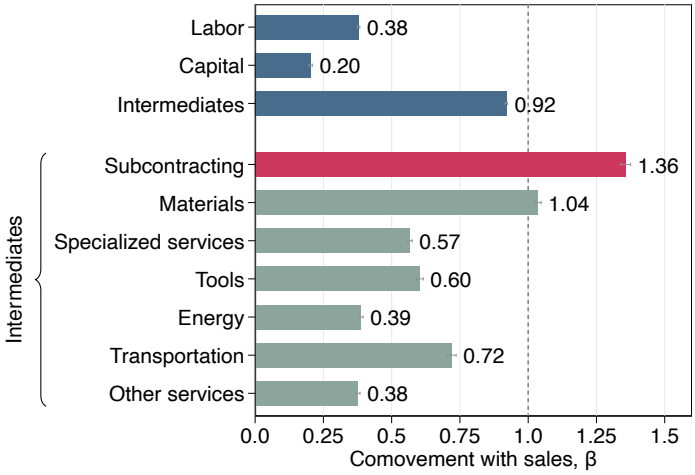
$$\Delta \log \text{input}_{it} = \beta \, \Delta \log \text{sales}_{it} + \text{sector} \times \text{year FE} + \epsilon_{it}$$



All sales changes | All firms | Subcategories | Linearity

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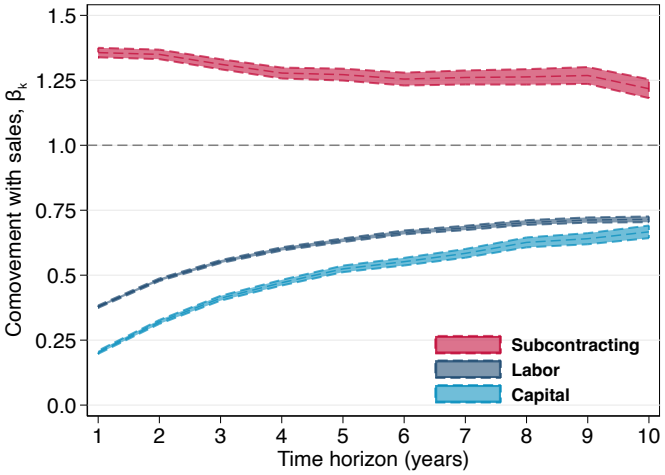
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FACT 4. Subcontracting comovement with sales falls at longer horizons

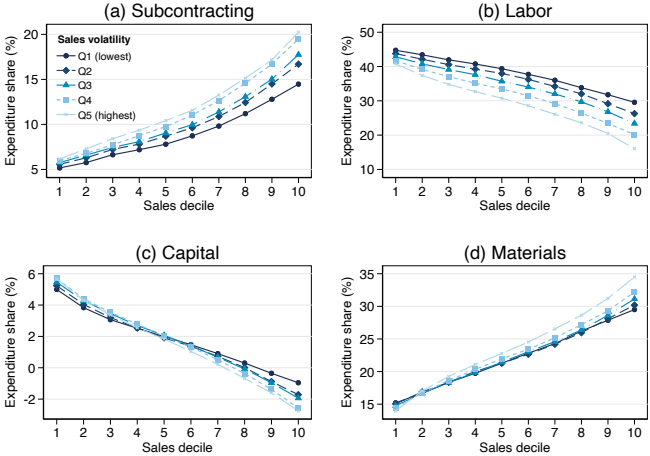
$$\Delta_k \log \text{input}_{it} = \beta_k \Delta_k \log \text{sales}_{it} + \text{sector} \times \text{year FE} + \epsilon_{it}, \quad k = 1, \ldots, 10$$



Balanced panel | All inputs | Sales dispersion

FACT 5. Firms presenting more volatile sales subcontract more

$$\text{input share}_{it} = \beta^{q}_{\text{sales decile}_{it}} + \text{sector} \times \text{year FE} + \epsilon_{it}, \quad q = 1, \ldots, 5$$



Footnote | Other intermediates

Taking stock

- Subcontracting is widespread in the economy
  - Larger firms spend proportionally more on subcontracting
- Subcontracting comoves with sales more than any other input
  - Its comovement weakens at longer horizons, while that of labor and capital strengthens
  - Its expenditure share is larger for firms presenting more volatile sales

- Consistent with subcontracting as a flexible margin to accommodate demand fluctuations...
- ...alongside the classic specialization gains from delegating tasks
- Next: a model to separate the two forces and measure their aggregate value

Empirics

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Environment – Technology

- Continuum of sequential tasks $s \in [0, 1]$: labor transforms materials m_{it} into physical output g_{it}
  Ciccone (2002); Costinot et al. (2013)

$$\frac{d \ln q_{it}(s)}{ds} = b[\ln A_{it}(s)\, \ell_{it}(s) - \ln q_{it}(s)]$$

$$q_{it}(0) = m_{it}, \quad g_{it} = q_{it}(1)$$



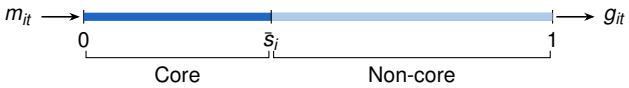
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- Core tasks ($s < \bar{s}_i$) in-house; non-core tasks ($s \geq \bar{s}_i$) can be subcontracted

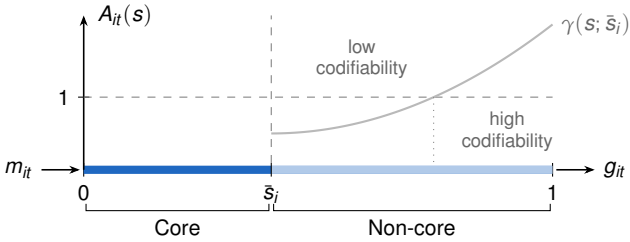
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- Task productivity: $A_{it}(s) = \begin{cases} 1 & \text{in-house} \\ \gamma(s; \bar{s}_i) & \text{subcontracted} \end{cases}$
  Acemoglu and Restrepo (2018, 2019)

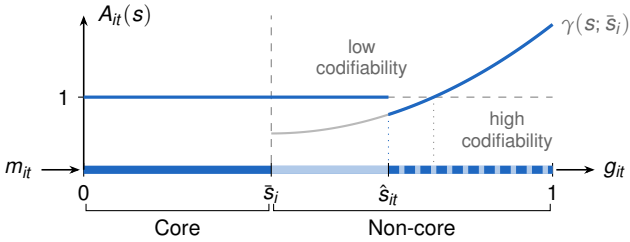
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- Firm chooses the subcontracting threshold $\hat{s}_{it}$

Types

Environment – Revenue, shocks and frictions

- Revenue: $y_{it} = z_{it}\,\varepsilon_{it}\,g_{it}^{\nu}, \quad \nu < 1$
- Fixed cost of operation: κ_F

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- Productivity: persistent and transitory

$$\log z_{it} = \rho \log z_{it-1} + \log \eta_{it}$$
$$\log \eta_{it} \sim \textit{iid}\ \mathcal{N}(0, \sigma_{\eta}^2)$$
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- In-house labor $l_{H,it} = \int_0^{\hat{s}_{it}} \ell_{H,it}(s)\, ds$:

$$c(l_{H,it}, l_{H,it-1}) = \begin{cases} w_t \frac{\psi_H}{2} \frac{(\Delta l_{H,it})^2}{l_{H,it-1}} & \text{if } \Delta l_{H,it} > 0 \quad \text{(hiring)} \\ w_t \frac{\psi_F}{2} \frac{(\Delta l_{H,it})^2}{l_{H,it-1}} & \text{if } \Delta l_{H,it} < 0 \quad \text{(firing)} \end{cases}$$

Environment – Subcontracting

- Representative, perfectly competitive subcontractor
- Subcontractor's efficiency on delegated task s: $\gamma(s; \bar{s})$
- Subcontractor hires workers at w_t, faces same adjustment costs on its total labor

Entry and timing

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- Access: labor effort e in adapting the technology succeeds w.p. $p(e)$
- Access obsolescence: lost w.p. δ each period

Entry and timing

Subcontractor's problem and pricing

- Subcontractor takes the price ω_t as given:

$$W_t(L_{S,t-1}) = \max_{L_{S,t} \geq 0} \left\{ (\omega_t - w_t) L_{S,t} - c(L_{S,t}, L_{S,t-1}) + \beta W_{t+1}(L_{S,t}) \right\}$$

- Optimality: price equals the wage plus the marginal cost of adjusting its workforce,

$$\omega_t = w_t + c_1(L_{S,t}, L_{S,t-1}) + \beta\, c_2(L_{S,t+1}, L_{S,t})$$

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- Pooling: firms' shocks are idiosyncratic, so some firms expand while others contract
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- Subcontractor never adjusts its workforce in steady state

$$\omega = w + \underbrace{c_1(L_S, L_S) + \beta\, c_2(L_S, L_S)}_{0} = w$$

Volatility in the data

Firms – problem with access

- States, $(z, \varepsilon, l_{H,-1})$: productivity and inherited labor
- Choices, $(\hat{\mu}, l_H, l_S, m)$, and exit: delegated weight, in-house and subcontracted labor, materials

$$V_{S,\bar{\mu}}(z, \varepsilon, l_{H,-1}) = \max_{\hat{\mu} \in [0, \bar{\mu}],\, l_H,\, l_S,\, m} \Big\{ y - w l_H - c(l_H, l_{H,-1}) - \omega l_S - m - \kappa_F + \beta \max \big\{ -c(0, l_H),\ \mathcal{V}_{S,\bar{\mu}}(z, l_H) \big\} \Big\}$$

$$y = z\varepsilon\, B(\hat{\mu})^{\nu} \Big[m^{\alpha_m} \Big(\frac{l_S}{\hat{\mu}} \Big)^{\hat{\mu}} \Big(\frac{l_H}{1 - \alpha_m - \hat{\mu}} \Big)^{1 - \alpha_m - \hat{\mu}} \Big]^{\nu}$$

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- $\hat{\mu}$: output weight of the delegated tasks $(\hat{s}, 1]$
- $B(\hat{\mu}) \equiv \exp\{ \int_0^{\hat{\mu}} \ln \tilde{\gamma}(u; \bar{\mu})\, du \}$: productivity shifter from the efficiency of delegated tasks

Without access | Technology

Firms – delegation when expanding and contracting

- Optimality condition on the delegated weight, $\hat{\mu}$:

$$\underbrace{\tilde{\gamma}(\hat{\mu};\bar{\mu})}_{\text{task efficiency}} = \underbrace{\frac{\omega}{\widetilde{w}}}_{\text{relative price}}$$

where

- ω: price per unit of the subcontractor's labor
- $\widetilde{w}$: shadow cost of in-house labor

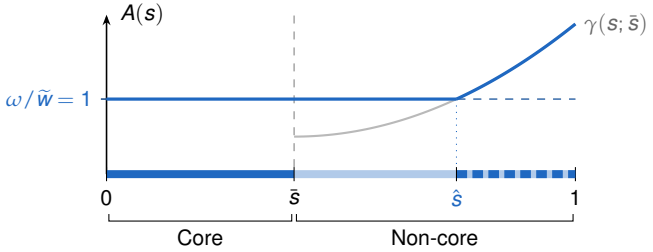
$$\widetilde{w} = \underbrace{w}_{\text{wage}} + \underbrace{c_1(l_H, l_{H,-1})}_{\text{adjusting today}} + \underbrace{\beta\,\Omega}_{\text{adjusting tomorrow}}$$

$$\Omega = \underbrace{\mathbf{1}\{\mathcal{V} \geq -c(0, l_H)\}\,\mathbb{E}\,c_2(l_{H,+1}, l_H)}_{\text{continue}} + \underbrace{\mathbf{1}\{\mathcal{V} < -c(0, l_H)\}\,c_2(0, l_H)}_{\text{exit}}$$

Policy functions | Types | Productivity shifter

Firms – delegation when expanding and contracting

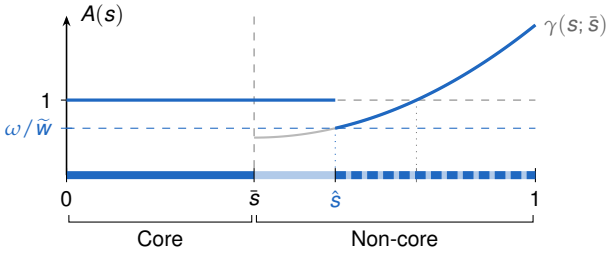
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- Constant size, $\widetilde{w} = \omega$: firm delegates tasks with efficiency above one

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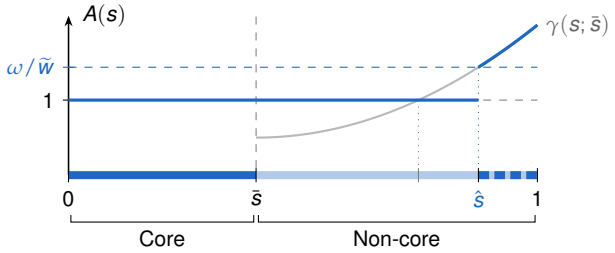


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- Expanding, $\widetilde{w} > \omega$: firm also delegates less codifiable tasks, to save hiring costs

Policy functions | Types | Productivity shifter

Firms – delegation when expanding and contracting

$$\tilde{\gamma}(\hat{\mu};\bar{\mu}) = \frac{\omega}{\widetilde{w}}$$



- Constant size, $\widetilde{w} = \omega$: firm delegates tasks with efficiency above one
- Expanding, $\widetilde{w} > \omega$: firm also delegates less codifiable tasks, to save hiring costs
- Contracting, $\widetilde{w} < \omega$: firm keeps more codifiable tasks in-house, to save firing costs

Policy functions | Types | Productivity shifter

Firms – access to subcontracting

- Firms without access choose labor effort e before observing productivity:

$$\mathcal{J}_{\bar{\mu}}(z, l_{H,-1}) = \max_{e \geq 0} \Big\{ -we + p(e)\, \mathbb{E}\big[V_{S,\bar{\mu}}(\cdot) - V_{NS,\bar{\mu}}(\cdot)\big] \Big\}$$

- Effort rises with the expected value of access, increasing in:

  → Subcontracting ability: delegation gains scale with the non-core segment, at any shadow cost

  → Productivity: delegation cuts unit costs proportionally, so it is worth more to more productive firms

  → Selection of firms into subcontracting

Aggregate productivity

The model admits an aggregate production function with aggregate productivity Z_t:

$$Y_t = Z_t \, \Lambda_t^{1-\nu} \left(M_t^{\alpha_m} \left(\frac{\mathcal{L}_t}{1-\alpha_m} \right)^{1-\alpha_m} \right)^{\nu}, \qquad Z_t = \mathcal{S}_t \, \mathcal{E}_t \, \mathcal{A}_t^{\nu}$$

- Mass of active firms, $\Lambda_t = \int_i di$
- Total production labor, $\mathcal{L}_t = \int_i \left(l_{H,it} + l_{S,it} \right) di$
- Materials, $M_t = \int_i m_{it} \, di$

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Selection:

$$\mathcal{S}_t \equiv \mathbb{E}_t[z_{it}\varepsilon_{it}]$$

- Mean productivity of active firms

Aggregate productivity

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Allocative efficiency:

$$\mathcal{E}_t \equiv \mathbb{E}_t \left[\frac{z_{it} \varepsilon_{it}}{\mathbb{E}_t [z_{it} \varepsilon_{it}]} \left(\frac{g_{it}}{\bar{g}_t} \right)^{\nu} \right]$$

- Physical output, $G_t = \int_i g_{it} \, di$, and its mean, $\bar{g}_t = G_t / \Lambda_t$
- Rises when firm size tracks firm productivity, falls with firm size dispersion

Aggregate productivity | Labor wedge

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Input efficiency:

$$\mathcal{A}_t \equiv \frac{G_t}{M_t^{\alpha_m} \left(\frac{\mathcal{L}_t}{1-\alpha_m} \right)^{1-\alpha_m}}$$

- Delegation gains raise input efficiency: cost savings on delegated tasks
- Dispersion in shadow costs lowers input efficiency

Empirics

Model

Calibration

Quantification

Calibration – strategy

- 18 parameters, matched jointly to 18 moments; 4 parameters set externally

Firm dynamics

- **Productivity shocks**: persistence of sales and dispersion of sales growth
- **Labor frictions**: comovement of the wage bill with sales
- Entry and exit: relative size of entrants and exiters, exit rates
- Returns to scale: profit share

Parameters set externally | Functional forms | Other parameters

Calibration – strategy

- 18 parameters, matched jointly to 18 moments; 4 parameters set externally

- Firm dynamics: productivity shocks, labor frictions, entry and exit, returns to scale

- Subcontracting

  - **Types**: delegable weight, drawn at entry, $\bar{\mu}_i/(1-\alpha_m-\bar{\mu}_i) \sim \log \mathcal{N}(m_\mu, s_\mu^2)$

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  - **Delegation gains**: subcontractor's efficiency defines the delegated share and its elasticity

$$\tilde{\gamma}(u;\bar{\mu}) = \phi\, e^{-(u/\bar{\mu})/v_0} \quad \Rightarrow \quad \frac{\hat{\mu}}{\bar{\mu}} = v_0\left[\ln\phi + \ln(\tilde{w}/\omega)\right] \quad \Rightarrow \quad \frac{\partial \ln \hat{\mu}}{\partial \ln(\tilde{w}/\omega)} = \frac{1}{\ln\phi + \ln(\tilde{w}/\omega)}$$

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    $\Rightarrow$ **Flexibility**, from changes: comovement of subcontracting pins down ϕ

    $\Rightarrow$ **Specialization**, from levels: subcontracting share and its rise with size pin down m_μ and v_0

Parameters set externally | Functional forms | Other parameters

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  - **Access**: labor effort e succeeds with probability

$$p(e) = 1 - \exp(-b_p\, e^{a_p})$$

    $\Rightarrow$ Start rate and share of firms subcontracting by size pin down b_p and a_p

    $\Rightarrow$ Access lost w.p. δ each period, pinned down by the continuation rate

Parameters set externally | Functional forms | Other parameters

Calibration – frictions and subcontracting

Parameter	Value	Target	Data	Model
Labor frictions				
Hiring cost, ψ_H	1.61	Comovement of wage bill, rising sales	0.42	0.42
Firing cost, ψ_F	2.61	Comovement of wage bill, falling sales	0.35	0.35
Delegation gains				
Subcontractor's efficiency, ϕ	1.83	Comovement of subcontracting	1.36	1.52
Average type, m_μ	−0.56	Subcontracting share of subcontractors	0.24	0.27
Type dispersion, s_μ	0.75	Between-firm variation, subc. share	0.69	0.74
Delegated share scale, v_0	1.36	Subc. share intensive margin, d10–d1	0.12	0.11
Access				
Access scale, b_p	0.192	Subcontracting start rate	0.12	0.12
Access curvature, a_p	0.096	Share of firms subcontracting, d10–d1	0.49	0.50
Probability of losing access, δ	0.127	Subcontracting continuation rate	0.85	0.87

Notes: cost shares over labor, materials and subcontracting, the inputs in the model.

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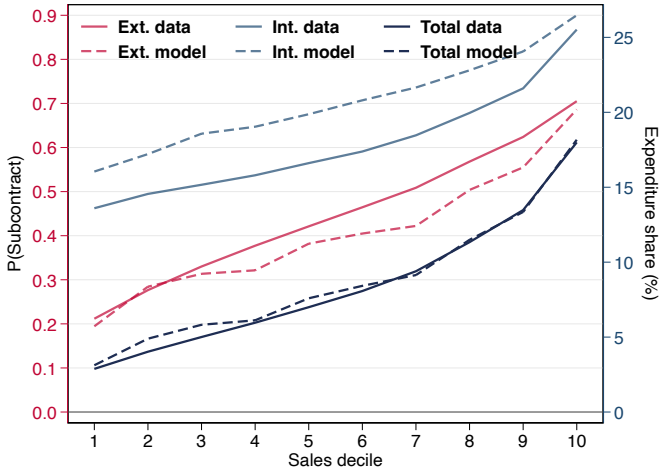
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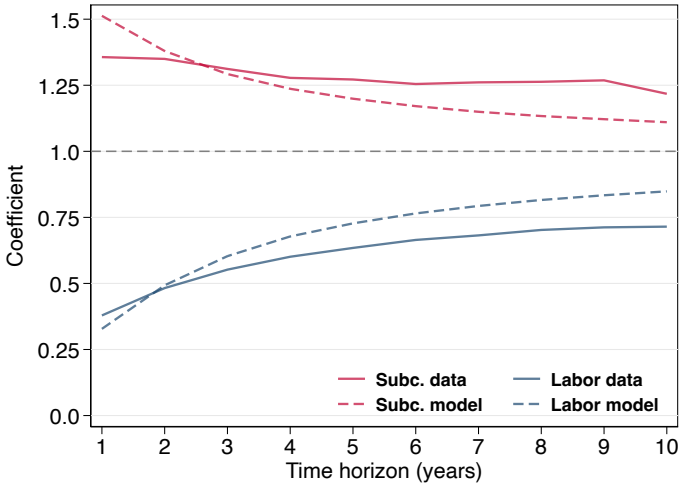
Calibration – size profile



Notes: the model lines compute FACT 2 shares on simulated firms. Cost shares over labor, materials and subcontracting, as in the model.

Firm size and subcontracting | Volatility | Size distribution and profits

Calibration – comovement across horizons



Notes: the model lines run the FACT 4 regression on simulated firms.

Empirics

Model

Calibration

Quantification

The gains from subcontracting

Q: What are the productivity gains from subcontracting?

- Exercise: solve the same economy without subcontracting

	Without subcontracting	With subcontracting
Productivity, Z	100.0	103.9
Selection, $\mathcal{S}$	100.0	100.2
Allocative efficiency, $\mathcal{E}$	100.0	99.9
Input efficiency, $\mathcal{A}^{\nu}$	100.0	103.8

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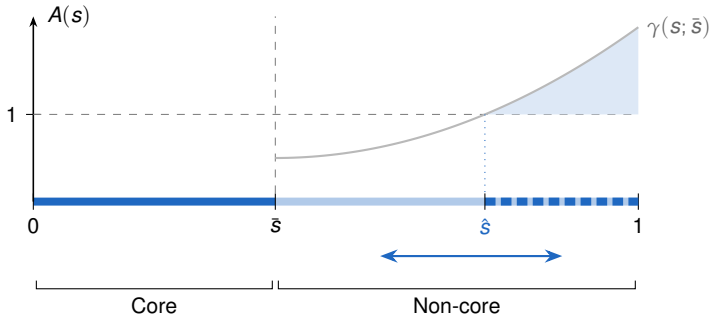
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$\Rightarrow$ Input efficiency increases and drives the productivity gains

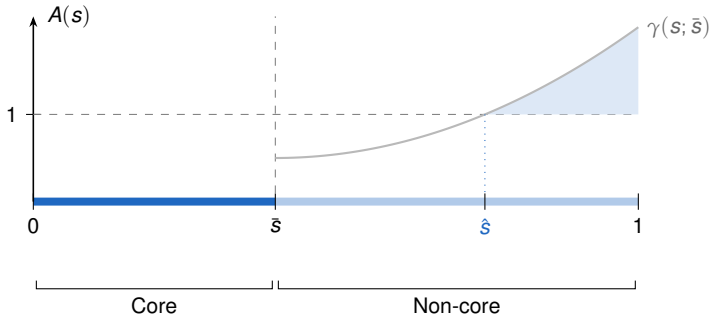
Flexibility versus specialization



With subcontracting

- Efficiency: gains from specialization
- Labor: delegated tasks subcontracted, threshold moves

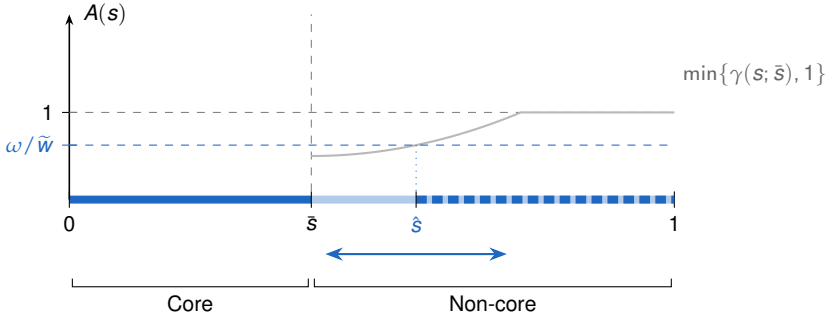
Flexibility versus specialization



Specialization only

- Efficiency: gains from specialization
- Labor: all in-house

Flexibility versus specialization



Flexibility only

- Efficiency: capped at one, no gains
- Labor: delegated tasks subcontracted, threshold moves

Decomposing the gains

Q: Which motive drives the productivity gains?

	Without subcontracting	Flexibility only	Specialization only	With subcontracting
Productivity, Z	100.0	100.5	102.9	103.9
Selection, $\mathcal{S}$	100.0	100.0	99.9	100.2
Allocative efficiency, $\mathcal{E}$	100.0	100.6	99.9	99.9
Input efficiency, $\mathcal{A}^{\nu}$	100.0	99.8	103.2	103.8
Labor wedge, τ_L	1.101	1.083	1.103	1.077
S.d. of sales growth	0.28	0.41	0.29	0.42
Corr. of physical output and prod.	0.80	0.83	0.79	0.79
Delegated task efficiency	–	0.99	1.35	1.34

Heterogeneity Types by size Subcontracting by size

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Heterogeneity | Types by size | Subcontracting by size

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$\Rightarrow$ Together: highest input efficiency, but specialization undoes the allocative efficiency gain from flexibility

Heterogeneity | Types by size | Subcontracting by size

The cost of adjustment frictions

Q: How costly are adjustment frictions when firms can subcontract?

- Without subcontracting: model recalibrated to the same data
- Exercise: in each economy, remove firing costs, hiring costs, or both

	Firing costs		Hiring costs		Both	
% change	With	Without	With	Without	With	Without
Productivity, Z	1.32	1.52	1.13	2.04	3.51	7.13
Selection, $\mathcal{S}$	1.11	0.59	0.14	0.22	0.40	0.59
Allocative efficiency, $\mathcal{E}$	0.34	0.80	0.72	1.74	2.84	6.02
Input efficiency, $\mathcal{A}^{\nu}$	–0.14	0.13	0.26	0.07	0.25	0.46

Recalibration

The cost of adjustment frictions

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$\Rightarrow$ Channel: allocative efficiency, since subcontracting already lets size track productivity

Recalibration

Aggregate versus idiosyncratic shocks

Q: Does flexibility survive aggregate shocks?

- Exercise: 1% productivity shock to one firm or to all firms
- Response: % change in the year of the shock, relative to no shock

	With subcontracting		Without subcontracting	
	Idiosyncratic	Aggregate	Idiosyncratic	Aggregate
Output	4.36	1.87	3.01	1.82
In-house labor	2.33	0.43	1.87	0.42
Subcontracted labor	6.01	0.43	–	–
Wage, w	0.00	0.87	0.00	0.86
Subcontracting price, ω	0.00	1.44	–	–

Metrics Transitions All responses Value of access

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Metrics Transitions All responses Value of access

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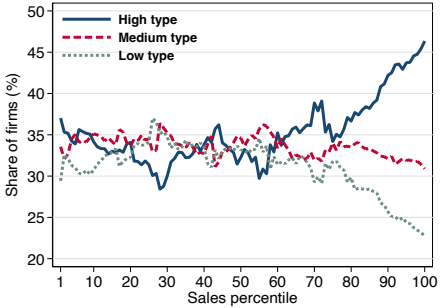
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$\Rightarrow$ Similar aggregate responses: imperfect pooling, adjustment costs pass through to firms

Metrics Transitions All responses Value of access

Subcontracting and the largest firms

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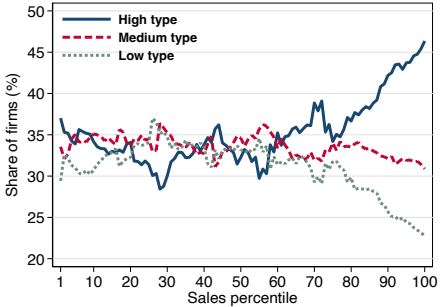


- Firms at the top are predominantly high types

All types | Firm size and subcontracting

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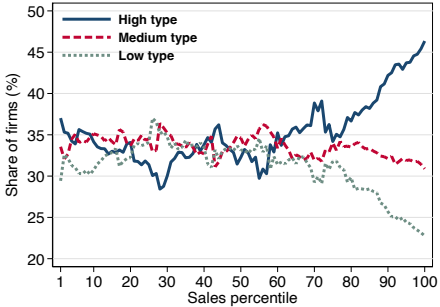


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  - ⇒ Specialization: high types gain more from delegation at every size

All types | Firm size and subcontracting

Subcontracting and the largest firms

Q: How does subcontracting shape who the largest firms are?

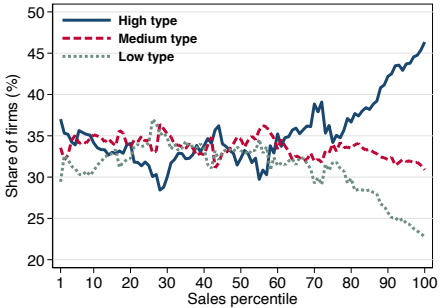


- Firms at the top are predominantly high types
  - ⇒ Specialization: high types gain more from delegation at every size
  - ⇒ Flexibility: high types can grow more through the subcontractor instead of hiring

All types | Firm size and subcontracting

Subcontracting and the largest firms

Q: How does subcontracting shape who the largest firms are?



Sales	Data	Baseline	No subc.
p99/p50	43.0	38.1	27.8
p95/p50	13.4	10.9	9.0
p90/p50	7.1	6.1	5.2
p75/p50	2.6	2.4	2.3
p50/p25	2.2	2.2	2.1
p50/p10	4.5	4.3	4.0
p50/p5	7.0	6.1	5.6

- Firms at the top are predominantly high types
  - ⇒ Specialization: high types gain more from delegation at every size
  - ⇒ Flexibility: high types can grow more through the subcontractor instead of hiring
- Without subcontracting, the top compresses much more than the bottom

All types | Firm size and subcontracting

Conclusion

- Firms delegate production **beyond their boundaries**
  - Large firms subcontract a sizeable share of production
  - Subcontracting tracks sales more than any other input
- Subcontracting's motives shape productivity differently
  - **Specialization**: efficiency in delegated tasks, **81%** of a **3.9%** gain
  - **Flexibility**: firms' size tracks productivity more closely, the other **19%**
- Measuring frictions requires the margins firms use to avoid them
  - Ignoring subcontracting **doubles** the productivity losses from adjustment frictions
  - Subcontracting shapes who the largest firms are: firms that can subcontract more are **over-represented**
- Under shock pooling, flexibility works for idiosyncratic shocks, much less for aggregate ones

Research agenda

The organization of production and aggregate productivity

Production Beyond the Firm: The Aggregate Value of Subcontracting this paper

Financial markets and firm dynamics

Financing the Scale-Up: Equity, Innovation and Economic Growth with Andrea Guccione

Does it matter for growth that debt and equity reach different firms?

The Role of Market Power in Capital Taxation

How does market power shape the choice between taxing capital income and wealth?

Innovation and growth

Leading or Following? Technology Adoption, Talent, and the Transatlantic Gap with Nicolò Ceneri

Is being the technological follower costly for Europe?

Appendix

Appendix

Back

Intermediate inputs definition

- *Materials*: goods intended to be physically incorporated into the final products
- *Subcontracting*: this account comprises the work necessary for the production process itself, for which the cooperation of other companies was obtained, subject to formal commitments or simple agreements
- *Specialized services*: technical services provided by other companies that the company itself cannot provide through its own resources, such as IT services, laboratory analyses, typographical work, studies, and expert opinions
- *Tools*: short-lived equipment, such as high-depreciation tools, office supplies, etc.
- *Energy*: energy and fluids, such as electricity, fuels or water
- *Transportation*: travel, accommodation and transportation of goods and materials
- *Other services*: rents, cleaning, communications, legal, insurance, IP

Appendix

Back

Summary statistics

	Mean	SD	p10	p25	p50	p75	p90
Sales	475	1,352	24	51	121	323	944
Profit	25	165	−32	−7	5	26	84
Profit share	5.2	21.0	−8.3	0.4	5.5	12.2	21.5
Labor cost	143	363	10	19	44	112	300
Capital cost	19	67	0	1	3	11	37
Intermediate inputs	288	958	10	21	57	171	539
Materials	114	525	0	0	0	37	181
Subcontracting	56	382	0	0	0	8	66
Employment	8	18	1	2	3	7	17
Exp. share: labor	31.8	19.0	11.0	17.6	27.6	41.8	59.9
Exp. share: capital	4.3	5.6	0.4	1.0	2.6	5.4	9.7
Exp. share: intermediates	63.9	19.5	34.8	52.5	67.6	78.8	86.3
Exp. share: materials	25.2	24.7	0.0	0.0	21.0	45.9	61.3
Exp. share: subcontracting	12.4	18.8	0.0	0.0	2.8	17.6	41.3

Notes: statistics computed for 1,758,186 firm-year observations. Ratios are weighted by the denominator variable. Monetary variables are expressed in thousands of 2022 EUR. Expenditure shares are expressed in percentages.

Appendix

Back

Sample selection

- **Economic activities** where
  - firms **sell what they produce**,
  - another firm can **carry out part of that production** on their behalf

$\Rightarrow$ Subcontracting has a clear meaning

- **Dropped** (CAE Rev. 3 divisions)
  - Wholesale and retail trade (45–47)
  - Accommodation and food services (55–56)
  - Real estate activities (68)
  - Rental and leasing activities (77)

- Results qualitatively identical if included

- **Observations**
  - 2010–2022: detailed input data start in 2010
  - No single-person firms (no balance sheet) or duplicates
  - Sales and costs non-missing and non-negative
  - Strictly positive labor and capital cost shares
  - At least one full-time paid worker

- **Trimming**, top and bottom
  - 0.5%: sales, each cost account, profits, expenditure and sales shares
  - 0.1%: one-year log changes in sales and in each variable cost

Appendix

Back

Subcontracting examples

- A **construction** firm subcontracts electrical installation to a specialized firm that brings its own workers and equipment
- A **garment** firm sends cut fabric to a workshop that stitches the garments and returns them finished
- A **consulting** firm subcontracts a data analysis module of a client project to another firm that delivers the finished report
- A private **hospital** subcontracts laboratory diagnostic services to an external lab that runs tests and returns results
- A **logistics** company subcontracts last-mile delivery in certain regions to local operators with their own vehicles and drivers

Appendix

Back

Input use (%)

	Share of total costs	Extensive margin	Intensive margin
Labor	31.8	100.0	31.8
Capital	4.3	100.0	4.3
Intermediates	63.9	100.0	63.9
Materials	25.2	48.8	36.6
Subcontracting	12.4	44.9	17.6
Specialized services	10.7	99.4	10.7
Tools	1.6	97.0	1.6
Energy	5.6	93.9	5.7
Transportation	2.7	81.9	2.8
Other services	5.8	99.2	5.8

Appendix

Back

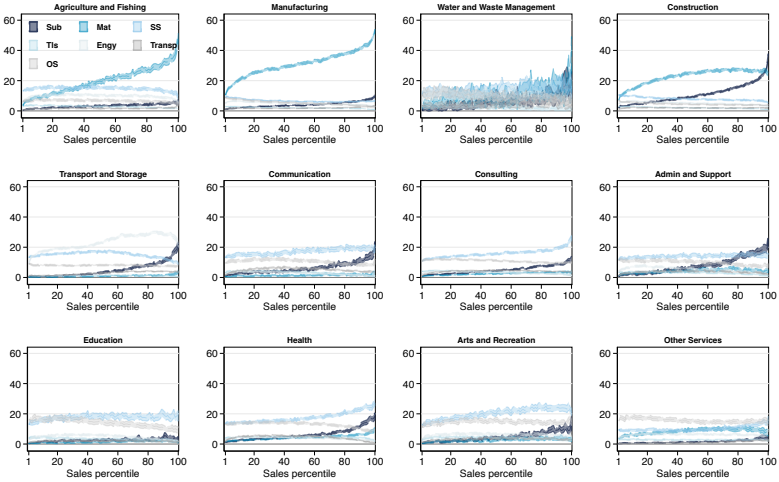
Subcontracting by sector (%)

	Share of total costs	Subcontracting share	Extensive margin	Intensive margin
Agriculture and Fishing	4.8	4.4	35.2	9.1
Mining and Quarrying	0.7	7.1	42.8	10.8
Manufacturing	41.1	7.1	52.8	10.0
Utilities	0.2	6.0	33.6	14.3
Water and Waste Management	1.1	12.3	48.7	18.4
Construction	17.0	24.3	64.6	27.6
Transport and Storage	8.8	19.2	34.1	24.3
Communication	4.0	13.9	41.1	21.4
Consulting	8.8	11.8	34.1	21.5
Admin and Support	4.0	18.0	42.9	27.8
Education	1.7	3.4	32.3	7.2
Health	5.4	12.8	41.0	20.3
Arts and Recreation	1.2	9.4	33.3	19.1
Other Services	1.1	3.6	29.0	7.7
All sectors	100.0	12.4	44.9	17.6

Appendix

Back

Firm size and intermediate input expenditures, by sector



Appendix

Back

Cost structure and firm size

- Input cost shares along the sales distribution:

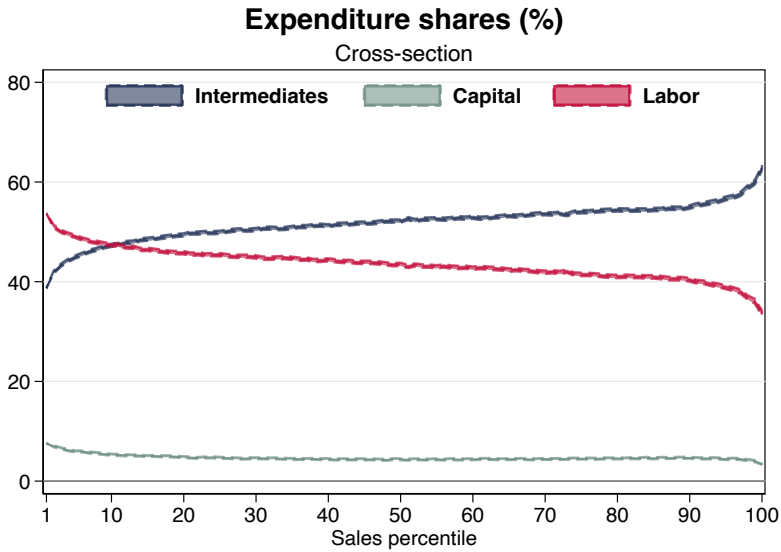
$$\text{input share}_{it} = \beta_{\text{sales percentile}_{it}} + \text{sector} \times \text{year FE} + \varepsilon_{it}$$

  - Input share: expenditure on the input over total costs
  - Sales percentile: within sector and year

Appendix

Back

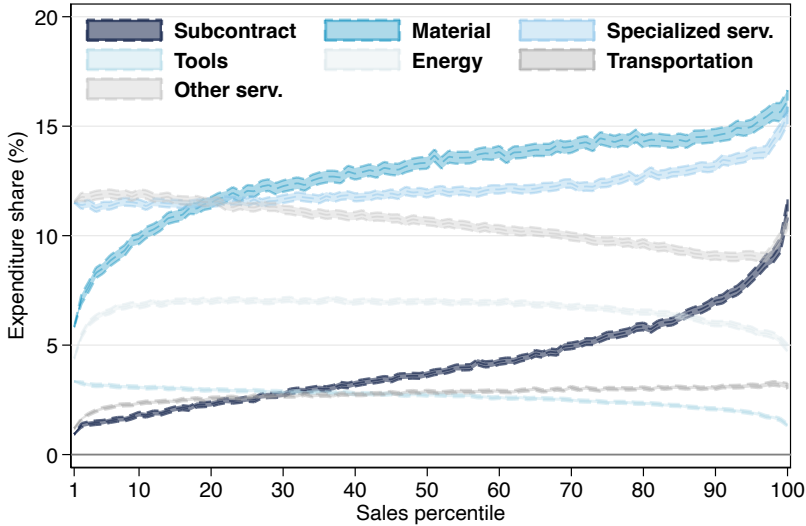
Firm size and main input expenditures



Appendix

Back

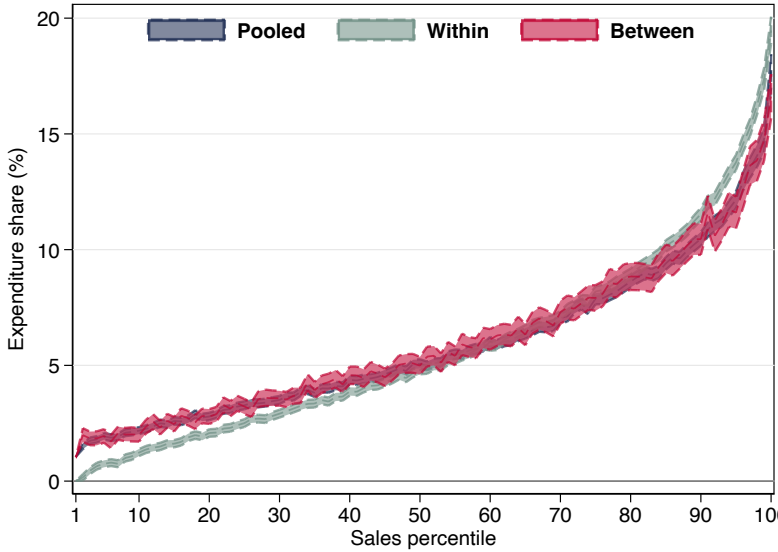
Firm size and intermediate input expenditures



Appendix

Back

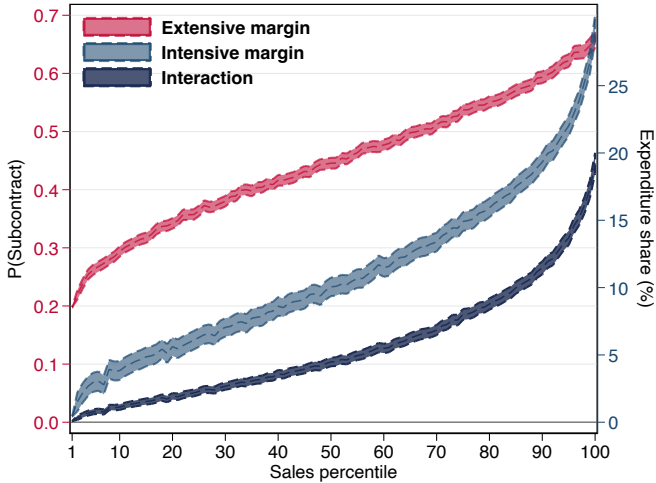
Between- vs within-firm



Appendix

Back

Within firm: firm and year fixed effects



Appendix

Back

Controlling for firm age: subcontracting share of total costs (p.p.)

	Without age	With age
Sales percentile, relative to the first		
20	1.81	1.98
40	3.36	3.57
60	4.97	5.25
80	7.69	8.08
100	16.95	17.58
Firm age in years, relative to 0–2		
3–5		−0.50
6–10		−0.92
11–15		−1.32
16–20		−1.37
More than 20		−1.65

Notes: sector × year FEs; firm-years with a known founding year (90%).

Appendix

Back

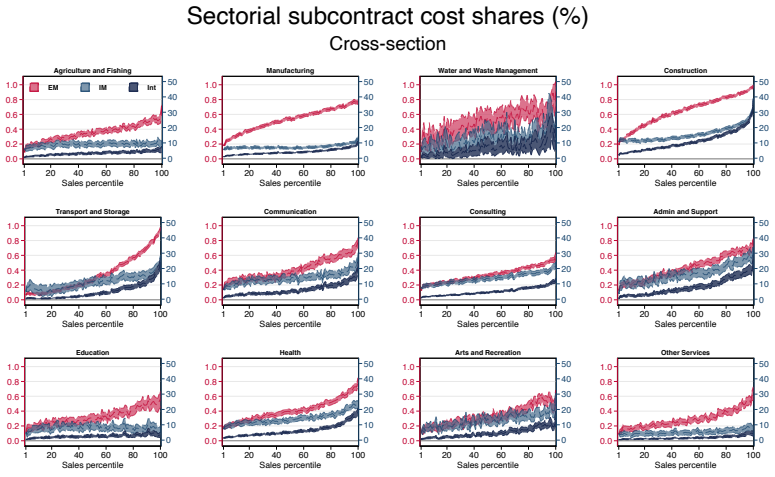
Other countries

- Larmat (2024):
  - French firms with 5+ employees, 2021
  - **38%** subcontract
  - Reasons: skills not available in-house (60%), reactivity to rising demand (50%)
- Fort (2017):
  - US manufacturing plants, 2007 Census of Manufactures
  - **29%** buy contract manufacturing (from other firms or plants)
  - These firms account for **71%** of manufacturing sales

Appendix

Back

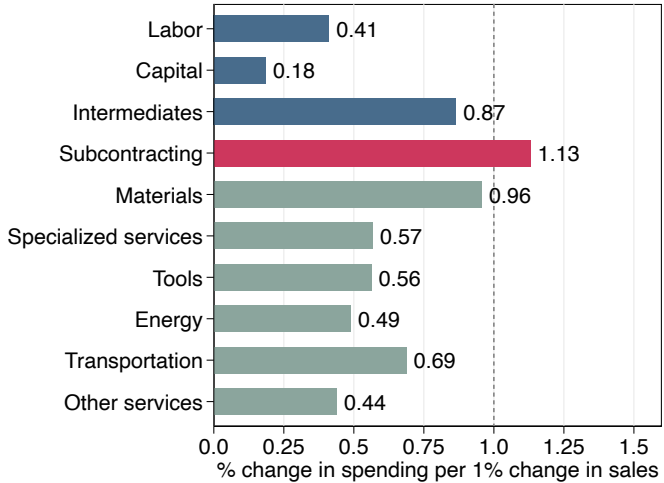
Both margins by sector



Appendix

Back

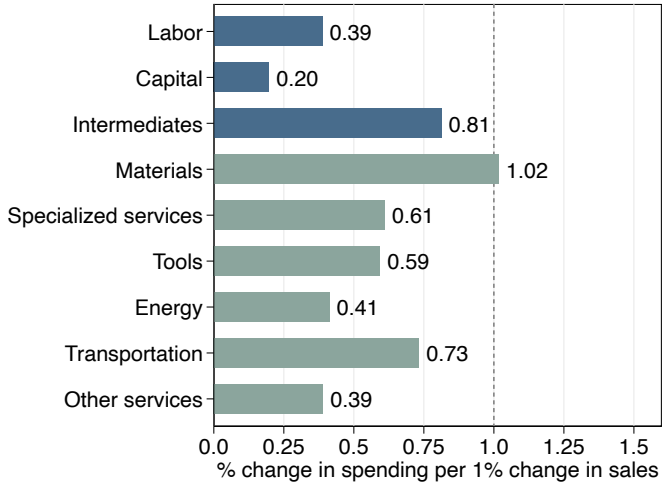
All sales changes



Appendix

Back

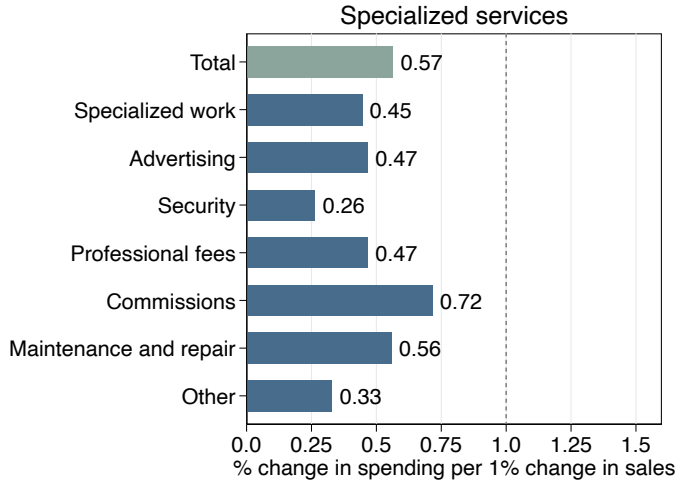
All firms



Appendix

Back

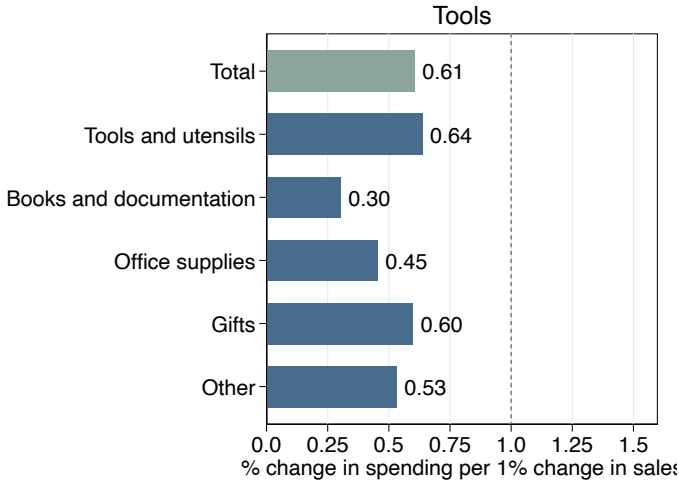
By subcategory



Appendix

Back

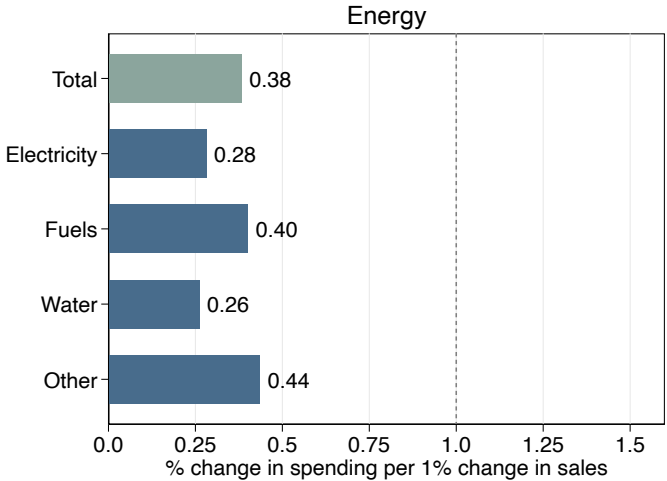
By subcategory



Appendix

Back

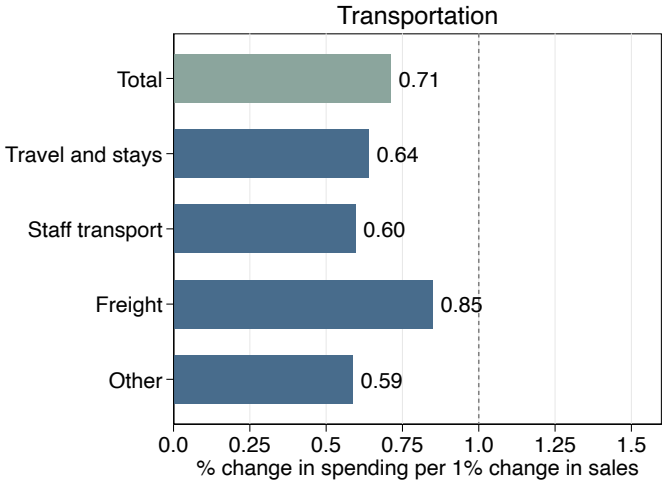
By subcategory



Appendix

Back

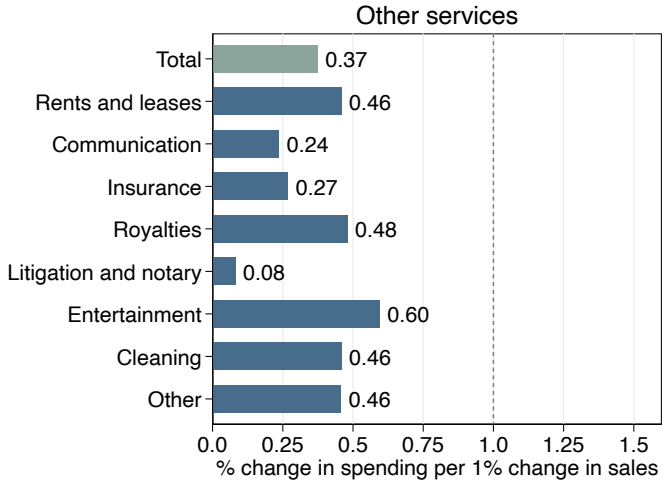
By subcategory



Appendix

Back

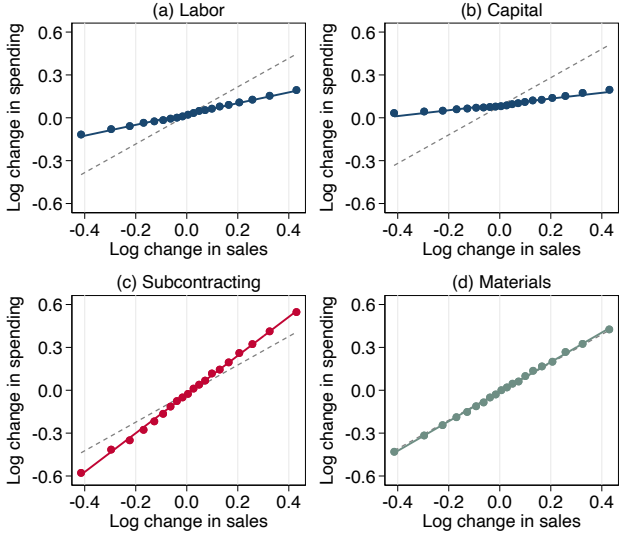
By subcategory



Appendix

Back

Inputs against sales changes, binned



Appendix

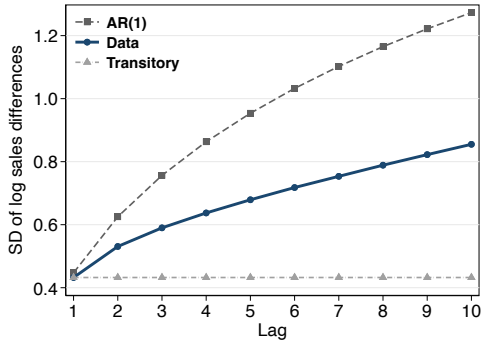
Back

Sales growth dispersion across horizons

- Persistent plus transitory process: $\ln y_{it} = p_{it} + \varepsilon_{it}$, $p_{it} = \rho\, p_{i,t-1} + \eta_{it}$
- Growth over k years, with $\sigma_p^2 = \sigma_\eta^2/(1-\rho^2)$:

$$\mathrm{Var}(\Delta_k \ln y_{it}) = \underbrace{2\sigma_p^2\,(1-\rho^k)}_{\text{persistent, rises with } k} + \underbrace{2\sigma_\varepsilon^2}_{\text{transitory, flat}}$$

$\Rightarrow$ Longer horizons accumulate persistent innovations: they over-represent the persistent component



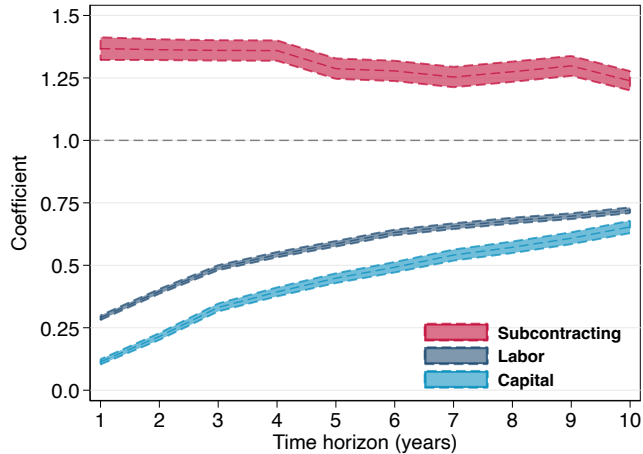
- Comovement with sales, β_k, a convex combination of the comovement with each component, β_k^p and β_k^ε:

$$\beta_k = \frac{\sigma_p^2(1-\rho^k)}{\sigma_p^2(1-\rho^k)+\sigma_\varepsilon^2}\,\beta_k^p + \frac{\sigma_\varepsilon^2}{\sigma_p^2(1-\rho^k)+\sigma_\varepsilon^2}\,\beta_k^\varepsilon$$

Appendix

Back

Comovement across horizons, balanced panel



Notes: firms observed in each of the ten previous years (2020–2022).

Appendix

Back

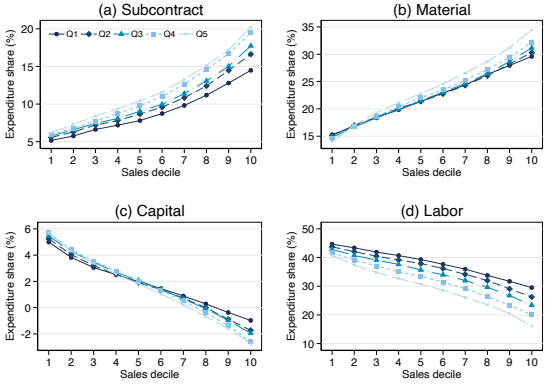
Comovement across horizons, all inputs

	Horizon (years)									
	1	2	3	4	5	6	7	8	9	10
Labor	0.379	0.482	0.552	0.601	0.634	0.664	0.682	0.702	0.712	0.715
	(0.002)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.006)
Capital	0.203	0.320	0.411	0.471	0.524	0.552	0.584	0.627	0.641	0.668
	(0.003)	(0.004)	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.010)	(0.011)	(0.012)
Intermediates	0.920	0.958	0.972	0.977	0.983	0.983	0.986	0.986	0.994	0.987
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Subcontracting	1.357	1.349	1.311	1.277	1.272	1.254	1.260	1.263	1.268	1.218
	(0.009)	(0.009)	(0.010)	(0.011)	(0.012)	(0.013)	(0.014)	(0.015)	(0.017)	(0.019)
Materials	1.035	1.045	1.056	1.054	1.064	1.072	1.071	1.078	1.099	1.081
	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)	(0.008)	(0.009)	(0.009)	(0.010)	(0.011)
Specialized services	0.566	0.639	0.680	0.701	0.711	0.719	0.735	0.739	0.752	0.751
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.007)	(0.008)	(0.009)
Tools	0.603	0.713	0.759	0.789	0.805	0.813	0.819	0.824	0.831	0.816
	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)	(0.008)	(0.009)	(0.010)	(0.011)	(0.012)
Energy	0.388	0.478	0.532	0.564	0.579	0.597	0.607	0.622	0.626	0.649
	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.007)	(0.008)	(0.009)
Transportation	0.719	0.828	0.896	0.936	0.951	0.970	0.976	0.979	0.973	0.989
	(0.008)	(0.009)	(0.009)	(0.010)	(0.012)	(0.013)	(0.014)	(0.015)	(0.017)	(0.019)
Other services	0.376	0.475	0.542	0.582	0.601	0.628	0.647	0.658	0.668	0.673
	(0.004)	(0.004)	(0.005)	(0.005)	(0.006)	(0.006)	(0.007)	(0.007)	(0.008)	(0.009)

Appendix

Back

Sales volatility, complete figure

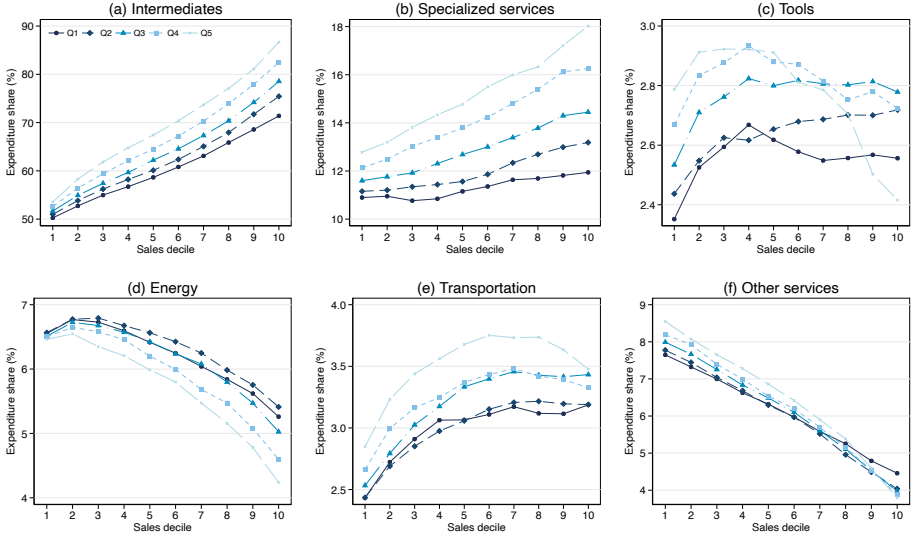


Notes: this figure shows expenditure shares for different inputs by recent firm-level volatility. Coefficients estimated from regressing expenditure shares on within-industry-year sales decile dummies for sales volatility quintiles separately. Results displayed for firms observed at least 5 years. Volatility is calculated as the standard deviation of the percentage change of sales over the last five periods: $\text{Vol}_{it} = \sqrt{\mathbb{V}_{t-5:t-1}\left(\frac{\text{pq}_{it}-\text{pq}_{it-1}}{\text{pq}_{it-1}}\right)}$. Q$x$ in the legend stands for the x-th quintile of the volatility measure distribution. Log of sales and sales growth rate are top and bottom trimmed at 1%. Standard errors clustered by firm (not shown) are significant at the 5% level for all regressions.

Appendix

Back

Sales volatility and other intermediates



Appendix

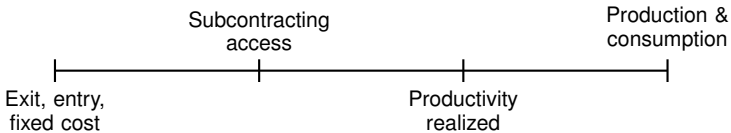
Back

Entry, preferences and timing

- Unbounded mass of potential entrants, E_t enters
- Upon paying κ_E, entrants draw type $\bar{\mu}$ from $F_{\bar{\mu}}$ and choose labor effort; then draw $\ln z \sim \mathcal{N}(\mu_E, \sigma_E^2)$ and ε
- Entrants start with zero workers; initial hiring without adjustment costs
- Representative household maximizes

$$U = \sum_{t=0}^{\infty} \beta^t \left(C_t - \theta \frac{N_t^{1+1/\iota}}{1+1/\iota} \right)$$

- Timing within the period:



Appendix

Back

Service sellers that do not subcontract are the least volatile

	Log s.d. of sales growth
Sells goods, does not subcontract	0.025
	(0.011)
Sells goods, subcontracts	0.242
	(0.014)
Sells services, subcontracts	0.167
	(0.019)
Log sales	−0.126
	(0.003)
Sector × year fixed effects	Yes
Obs.	212,180
Firms	30,115

Notes: standard errors in parentheses, clustered by firm. Manufacturing firms (CAE codes 10 to 32) observed at least five years. Dependent variable: log standard deviation of the annual percentage change in sales over the five previous years. Coefficients are relative to firms that sell services and do not subcontract. A firm sells services if services are at least 90% of its sales, and subcontracts if subcontracting is at least 5% of its total costs.

Appendix

Back

Problem without access

$$V_{NS,\bar{\mu}}(z,\varepsilon,l_{H,-1}) = \max_{l_H,\,m}\Big\{y - wl_H - c(l_H,l_{H,-1}) - m - \kappa_F + \beta\max\big\{-c(0,l_H),\ \mathcal{V}_{NS,\bar{\mu}}(z,l_H)\big\}\Big\}$$

$$y = z\varepsilon\Big[m^{\alpha_m}\Big(\frac{l_H}{1-\alpha_m}\Big)^{1-\alpha_m}\Big]^{\nu}$$

Appendix

Back

From stages to weights

- Production line: stages $s \in [0,1]$, materials enter at $s = 0$

$$\ln g = \alpha_m \ln m + \int_0^1 \zeta(s) \ln\left(A(s)\,\ell(s)\right) ds$$

with stage weights $\zeta(s) = b\, e^{-b(1-s)}$, $b = -\ln \alpha_m$

- Change of variables: **output weight downstream of s**

$$u(s) = \int_s^1 \zeta(t)\, dt = 1 - e^{-b(1-s)} \in [0,\ 1-\alpha_m]$$

- In weights, tasks enter symmetrically:

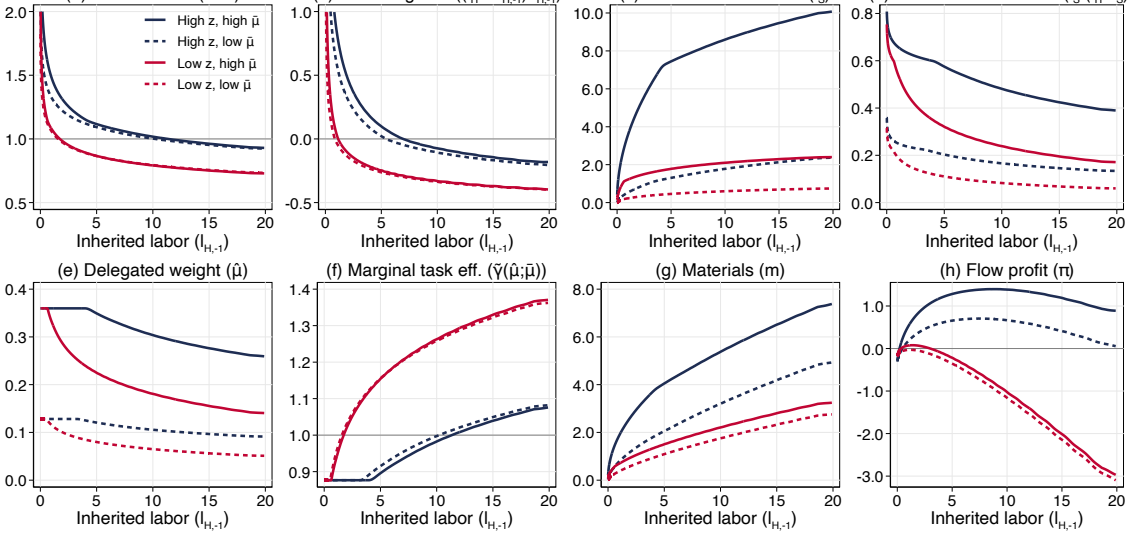
$$\ln g = \mathcal{K} + \alpha_m \ln m + \int_0^{1-\alpha_m} \ln\left(\widetilde{A}(u)\,\widetilde{\ell}(u)\right) du$$

- Delegated tasks $(\hat{s}, 1] \mapsto [0, \hat{\mu})$, $\hat{\mu} = 1 - e^{-b(1-\hat{s})}$, at productivity $\tilde{\gamma}(u; \bar{\mu})$

Appendix

Back

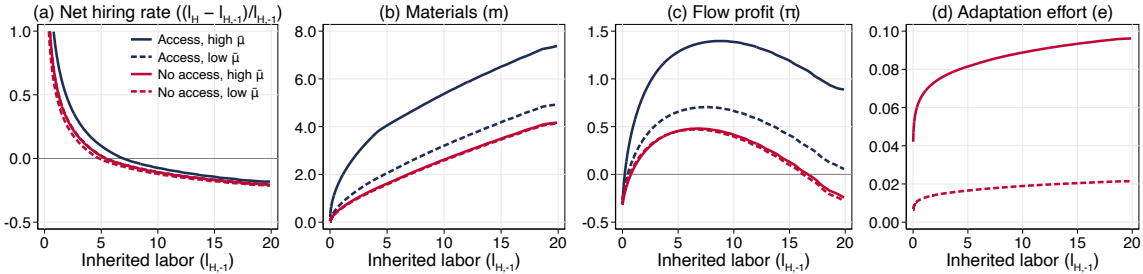
Policy functions – I



Appendix

Back

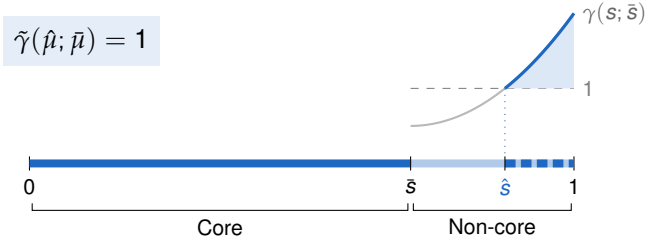
Policy functions – II



Appendix

Back to technology Back to delegation

Low type at constant size, $\bar{w} = \omega$

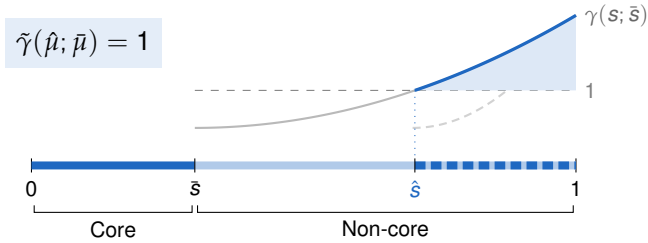


- Same efficiency profile, stretched over each type's non-core segment

Appendix

Back to technology Back to delegation

High type at constant size, $\bar{w} = \omega$

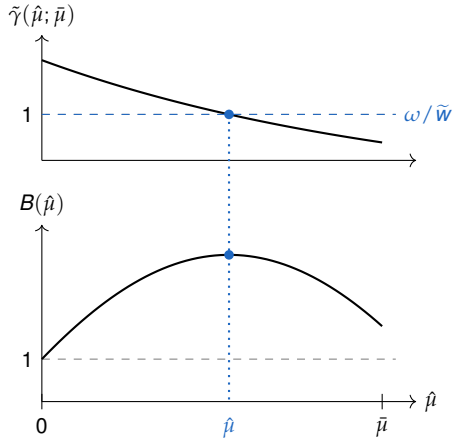


- Same efficiency profile, stretched over each type's non-core segment
- Same share delegated at any $\tilde{w}/\omega$, so the gain grows with the segment

Appendix

Back

Firms trade specialization for flexibility



$$B(\hat{\mu}) \equiv \exp\left\{\int_0^{\hat{\mu}} \ln \tilde{\gamma}(u;\bar{\mu})\, du\right\}$$

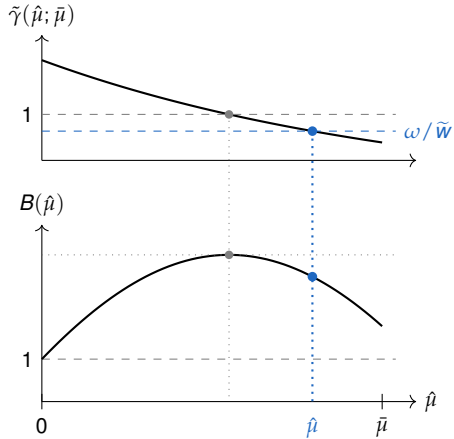
$$\frac{d \ln B(\hat{\mu})}{d\hat{\mu}} = \ln \tilde{\gamma}(\hat{\mu};\bar{\mu})$$

- B rises where $\tilde{\gamma} > 1$, falls where $\tilde{\gamma} < 1$
- Constant size, $\tilde{w} = \omega$: peak of B

Appendix

Back

Firms trade specialization for flexibility



$$B(\hat{\mu}) \equiv \exp\left\{ \int_0^{\hat{\mu}} \ln \tilde{\gamma}(u;\bar{\mu})\, du \right\}$$

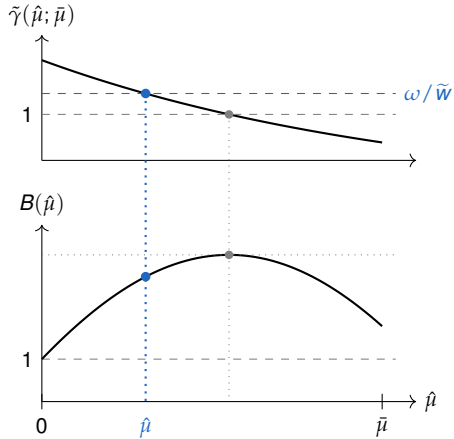
$$\frac{d \ln B(\hat{\mu})}{d\hat{\mu}} = \ln \tilde{\gamma}(\hat{\mu};\bar{\mu})$$

- B rises where $\tilde{\gamma} > 1$, falls where $\tilde{\gamma} < 1$
- Constant size, $\tilde{w} = \omega$: peak of B
- Expanding, $\tilde{w} > \omega$: past the peak

Appendix

Back

Firms trade specialization for flexibility



$$B(\hat{\mu}) \equiv \exp\left\{\int_0^{\hat{\mu}} \ln \tilde{\gamma}(u;\bar{\mu})\, du\right\}$$

$$\frac{d \ln B(\hat{\mu})}{d\hat{\mu}} = \ln \tilde{\gamma}(\hat{\mu};\bar{\mu})$$

- B rises where $\tilde{\gamma} > 1$, falls where $\tilde{\gamma} < 1$
- Constant size, $\tilde{w} = \omega$: peak of B
- Expanding, $\tilde{w} > \omega$: past the peak
- Contracting, $\tilde{w} < \omega$: short of the peak

Appendix

Back

Firm entry

- Free entry: firms enter until the marginal entrant breaks even,

$$\int \max_{e \geq 0} \Big\{ -we + p(e)\, \mathbb{E}_{\hat{G}}\big[V_{S,\bar{\mu}}(z,\varepsilon,0)\big] + \big(1-p(e)\big)\, \mathbb{E}_{\hat{G}}\big[V_{NS,\bar{\mu}}(z,\varepsilon,0)\big] \Big\}\, dF_{\bar{\mu}}(\bar{\mu}) \;\leq\; \kappa_E,$$

  with equality whenever the mass of entrants $E > 0$

- Entrants draw type $\bar{\mu}$ from $F_{\bar{\mu}}$ and choose labor effort; then draw (z, ε) from the entrants' distribution $\hat{G}$, start with zero workers and hire without adjustment costs

Appendix

Back

Stationary equilibrium – I

A stationary equilibrium consists of allocations $\{C, N\}$, prices $\{w, \omega\}$, policies $\{l_H, m, \hat{\mu}, l_S, e, \chi, e_E\}$, values $\{V_{NS}, V_S, \mathcal{V}_{NS}, \mathcal{V}_S, \mathcal{J}\}$, a mass of entrants E, and a distribution $\lambda(z, \varepsilon, l_{H,-1}, \bar{\mu}, a)$ such that:

1. Given prices, policies solve the firm problems V_{NS}, V_S, the effort problem $\mathcal{J}$, entrants' effort, and the exit decision
2. The subcontracting price satisfies the optimality condition, implying $\omega = w$ in steady state
3. Allocations maximize household utility given prices (labor supply and budget constraint)
4. The free entry condition holds, with equality if $E > 0$:

$$\int \max_{e \geq 0} \left\{ -we + p(e)\, \mathbb{E}_{\hat{G}}\left[V_{S,\bar{\mu}}(z, \varepsilon, 0)\right] + \left(1 - p(e)\right) \mathbb{E}_{\hat{G}}\left[V_{NS,\bar{\mu}}(z, \varepsilon, 0)\right] \right\} dF_{\bar{\mu}}(\bar{\mu}) \leq \kappa_E$$

Notation: $x = (z, \varepsilon, l_{H,-1}, \bar{\mu})$, $\widetilde{\chi}(x, a) = \chi(z, l_H(x, a), \bar{\mu}, a)$, $\tilde{e}(x) = e(z, l_H(x, 0), \bar{\mu})$

5. Labor market clears, with $L_S = \int l_S(x)\, d\lambda(x, 1)$:

$$N = L_S + \int l_H(x, a)\, d\lambda(x, a) + \int \left(1 - \widetilde{\chi}(x, 0)\right) \tilde{e}(x)\, d\lambda(x, 0) + E \int e_E(\bar{\mu})\, dF_{\bar{\mu}}(\bar{\mu})$$

6. Goods market clears, with $\Lambda = \int d\lambda(x, a)$ the mass of active firms:

$$Y - C - \int m\, d\lambda = \int w \tfrac{\psi_H}{2} \frac{(l_H - l_{H,-1})^2}{l_{H,-1}} \mathbf{1}\{l_H > l_{H,-1} > 0\}\, d\lambda + \Lambda \kappa_F + E \kappa_E$$

7. The distribution is consistent with policies, entry, and exit. With $\lambda'_a = \lambda(z', \varepsilon', l', \bar{\mu}', a)$ and $F' = F(z', \varepsilon'|z)$:

$$\begin{aligned}
\lambda'_1 = &\int_{\bar{\mu}\le\bar{\mu}'} (1-\widetilde{\chi}(x,1))(1-\delta)\,\mathbf{1}\{l_H(x,1)\le l'\}\,F'\,d\lambda(x,1) \\
&+ \int_{\bar{\mu}\le\bar{\mu}'} (1-\widetilde{\chi}(x,0))\,p(\tilde{e}(x))\,\mathbf{1}\{l_H(x,0)\le l'\}\,F'\,d\lambda(x,0) \\
&+ E\,\hat{G}(z',\varepsilon') \int_{\bar{\mu}\le\bar{\mu}'} p(e_E(\bar{\mu}))\,dF_{\bar{\mu}}(\bar{\mu}) \\
\lambda'_0 = &\int_{\bar{\mu}\le\bar{\mu}'} (1-\widetilde{\chi}(x,1))\,\delta\,\mathbf{1}\{l_H(x,1)\le l'\}\,F'\,d\lambda(x,1) \\
&+ \int_{\bar{\mu}\le\bar{\mu}'} (1-\widetilde{\chi}(x,0))\,(1-p(\tilde{e}(x)))\,\mathbf{1}\{l_H(x,0)\le l'\}\,F'\,d\lambda(x,0) \\
&+ E\,\hat{G}(z',\varepsilon') \int_{\bar{\mu}\le\bar{\mu}'} (1-p(e_E(\bar{\mu})))\,dF_{\bar{\mu}}(\bar{\mu})
\end{aligned}$$

Appendix

Back

Continuation values

$$\mathcal{V}_{NS,\bar{\mu}}(z, l_{H,-1}) = \mathbb{E}\left[V_{NS,\bar{\mu}}(\cdot)\right] + \mathcal{J}_{\bar{\mu}}(z, l_{H,-1})$$
$$\mathcal{V}_{S,\bar{\mu}}(z, l_{H,-1}) = (1-\delta)\,\mathbb{E}\left[V_{S,\bar{\mu}}(\cdot)\right] + \delta\,\mathbb{E}\left[V_{NS,\bar{\mu}}(\cdot)\right]$$

Appendix

Back

Delegation lowers unit costs

- At an interior optimum, the unit cost of a firm with access is

$$c_g = e^{-R(\hat{\mu})}\,\alpha_m^{-\alpha_m}\,\widetilde{w}^{\,1-\alpha_m}, \qquad R(\hat{\mu}) = \int_0^{\hat{\mu}} \ln \frac{\tilde{\gamma}(u;\bar{\mu})}{\tilde{\gamma}(\hat{\mu};\bar{\mu})}\,du \geq 0, \qquad \frac{\partial \ln c_g}{\partial \ln \widetilde{w}} = 1 - \alpha_m - \hat{\mu}$$

- Productivity: revenue is $y = (z\varepsilon)^{\frac{1}{1-\nu}}(\nu/c_g)^{\frac{\nu}{1-\nu}}$, so a proportional cut in c_g is worth more at higher $z\varepsilon$
- Subcontracting ability:
  - Types share one efficiency profile, stretched over their non-core segment
  - Delegation gains, $R(\hat{\mu})$, scale with the segment at any shadow cost

Appendix

Back

Aggregate productivity

Aggregates: $Y_t = \int_i z_{it}\varepsilon_{it}\, g_{it}^{\nu}\, di, \quad G_t = \int_i g_{it}\, di, \quad M_t = \int_i m_{it}\, di$

$$\mathcal{L}_t = \int_i \left(l_{H,it} + l_{S,it}\right) di, \quad \Lambda_t = \int_i di, \quad \bar{g}_t = \frac{G_t}{\Lambda_t}$$

Productivity: $$Z_t \equiv \frac{Y_t}{\Lambda_t^{1-\nu}\left(M_t^{\alpha_m}\left(\frac{\mathcal{L}_t}{1-\alpha_m}\right)^{1-\alpha_m}\right)^{\nu}} = \mathcal{S}_t\, \mathcal{E}_t\, \mathcal{A}_t^{\nu}$$

Selection: $$\mathcal{S}_t \equiv \mathbb{E}_t[z_{it}\varepsilon_{it}]$$

Allocative efficiency: $$\mathcal{E}_t \equiv \mathbb{E}_t\left[\frac{z_{it}\varepsilon_{it}}{\mathbb{E}_t[z_{it}\varepsilon_{it}]}\left(\frac{g_{it}}{\bar{g}_t}\right)^{\nu}\right] = \frac{Y_t}{\Lambda_t^{1-\nu} G_t^{\nu}\, \mathbb{E}_t[z_{it}\varepsilon_{it}]}$$

Input efficiency: $$\mathcal{A}_t \equiv \frac{G_t}{M_t^{\alpha_m}\left(\frac{\mathcal{L}_t}{1-\alpha_m}\right)^{1-\alpha_m}} = \frac{\mathbb{E}_{\varsigma}\left[e^{R(\hat{\mu}_{it})}\, \widetilde{w}_{it}^{\alpha_m}\right]}{\left(\mathbb{E}_{\varsigma}[\widetilde{w}_{it}]\right)^{\alpha_m}\left(\mathbb{E}_{\varsigma}[h_{it}]\right)^{1-\alpha_m}}$$

$$h_{it} = 1 + \frac{\hat{\mu}_{it}}{1-\alpha_m}\left(\frac{\widetilde{w}_{it}}{\omega_t} - 1\right), \quad \varsigma_{it} \propto \frac{l_{H,it}}{1-\alpha_m-\hat{\mu}_{it}}$$

$$R(\hat{\mu}) = \int_0^{\hat{\mu}} \ln \frac{\tilde{\gamma}(u;\bar{\mu})}{\tilde{\gamma}(\hat{\mu};\bar{\mu})}\, du$$

If $\hat{\mu}_{it} = 0$: $$\mathcal{A}_t = \frac{\mathbb{E}_{\varsigma}\left[\widetilde{w}_{it}^{\alpha_m}\right]}{\left(\mathbb{E}_{\varsigma}[\widetilde{w}_{it}]\right)^{\alpha_m}} \leq 1$$

Appendix

Back

Aggregate labor wedge

Firm level: $\widetilde{w}_{it}\, l_{H,it} = (1 - \alpha_m - \hat{\mu}_{it})\, \nu y_{it}, \quad \omega_t\, l_{S,it} = \hat{\mu}_{it}\, \nu y_{it}$

Wedge: $\nu(1-\alpha_m)\dfrac{Y_t}{\mathcal{L}_t} = \tau_{L,t}\, w_t$

$$\tau_{L,t} = \frac{\bar{w}_t}{w_t}\, s_{H,t} + \frac{\omega_t}{w_t}\,(1 - s_{H,t})$$

Components: $\bar{w}_t = \displaystyle\int_i \widetilde{w}_{it}\, \frac{l_{H,it}}{L_{H,t}}\, di, \quad s_{H,t} = \frac{L_{H,t}}{\mathcal{L}_t}$

$$L_{H,t} = \int_i l_{H,it}\, di$$

If $\omega_t = w_t$: $\tau_{L,t} = 1 + \left(\dfrac{\bar{w}_t}{w_t} - 1\right) s_{H,t}$

If $\widetilde{w}_{it} = w_t$: $\tau_{L,t} = 1$

Appendix

Back

Calibration – functional forms

- Access probability: $p(e) = 1 - \exp\big(-b_p\, e^{a_p}\big), \quad a_p \in (0,1),\ b_p > 0$
- Efficiency profile: $\tilde{\gamma}(u;\bar{\mu}) = \phi\, e^{-v/v_0}, \quad v = u/\bar{\mu} \ \Rightarrow\ \hat{\mu} = \bar{\mu}\min\big\{v_0 \max\big\{\ln\phi + \ln(\widetilde{w}/\omega),\ 0\big\},\ 1\big\}$
- Firm types: $\bar{\mu}/(1-\alpha_m-\bar{\mu}) \sim \log\mathcal{N}(m_\mu, s_\mu^2),\ N = 3$

Appendix

Back

Calibration – parameters set externally

Parameter	**Value**	**Source**
Discount factor, β	0.978	Real interest rate of 2.3%
Labor disutility scale, θ	1.0	Normalization
Frisch elasticity, ι	0.5	Standard value
Materials weight in variable inputs, α_m	0.363	Data

Appendix

Back

Calibration – other parameters

Parameter	Value	Target	Data	Model
Productivity shocks				
Persistence of z, ρ	0.983	Autocorrelation of log sales, 1 yr	0.95	0.95
S.d. of transitory shocks, σ_ε	0.078	S.d. of sales growth, 1 yr	0.43	0.42
S.d. of persistent innovations, σ_η	0.044	S.d. of sales growth, 3 yr	0.59	0.66
Entry and exit				
Mean entrant productivity, μ_E	0.179	Entrants' relative size	0.49	0.50
Dispersion of entrant productivity, σ_E	0.100	Entrants' exit rate rel. to incumbents	1.52	1.51
Exogenous exit probability, δ_X	0.060	Firm exit rate	0.08	0.08
Fixed cost, κ_F	0.165	Exiters' relative size	0.78	0.78
Entry cost, κ_E	6.67	Mean employment	7.77	7.77
Returns to scale				
Returns to scale, ν	0.908	Variable-cost share	0.92	0.88

Appendix

Back

Type composition with and without adjustment frictions

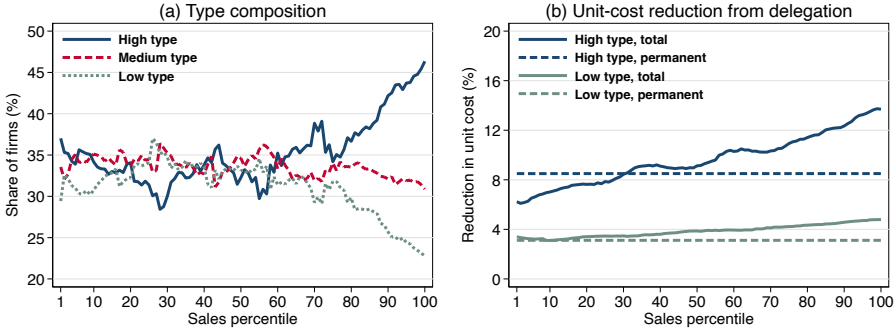
	With frictions			Without frictions		
	Low	Medium	High	Low	Medium	High
Below median	33.2	33.7	33.1	28.4	36.0	35.7
50th–90th percentile	30.3	33.7	36.0	28.8	36.7	34.5
Top 10%	23.5	31.8	44.8	28.9	30.5	40.6
Top 1%	18.8	29.3	51.9	27.3	31.1	41.6
All firms	31.0	33.5	35.4	28.6	35.7	35.7

Notes: share of each type among the active firms in each group, in percent, in the stationary distribution. Groups are defined by sales and hold exactly the stated share of firms; firms tied at a cutoff are split pro rata. Without frictions, $\psi_H = \psi_F = 0$, with prices in general equilibrium.

Appendix

Back

Calibration – firm size and subcontracting



- High types concentrate at the top: they gain more from delegation
  - Delegation saves 4% of unit costs for low types, 11% for high types
  - Specialization: same gap between types at every size (dashed lines)
  - Flexibility: gap widens from 3 to 9 p.p. with size, as large firms are hiring

Appendix

Back

Calibration – input usage and volatility

	Subcontracting		Labor		Materials	
	Data	Model	Data	Model	Data	Model
Sales volatility (std.)	1.02	3.67	−1.16	−3.54	0.14	−0.14
Log sales (std.)	4.98	3.22	−9.58	−4.03	4.61	0.81
Volatility × size	0.69	0.45	−0.77	−0.42	0.08	−0.03

Notes: dependent variables are expenditure shares in percentage points. Volatility is the standard deviation of the percentage change in sales over the previous five years. Volatility and log sales are standardized within each sample. Empirical cost shares have been recomputed by only including as costs the accounts present in the model: labor, materials and subcontracting. Data regressions absorb sector × year fixed effects; model regressions absorb year fixed effects only, since the model has no sectors. All coefficients except the interaction term for materials in the data are significant at the 1% level (standard errors clustered at the firm level).

Appendix

Back

Calibration – size distribution and profits

	Data	Model
Firms by number of workers (%)		
Below 5	63.3	67.9
5 to 9	18.7	16.1
10 to 19	9.9	9.3
20 to 49	5.9	4.8
50 to 99	1.6	1.3
100 or more	0.7	0.7
Firms with negative profits (%)	37.0	36.0
Sales, ratio of percentiles		
p90/p10	38.5	26.0
p90/p50	7.8	6.0
p50/p10	4.9	4.3

Appendix

Back

Delegation across the shadow cost of labor

Quintile of $\widetilde{w}$	Q1	Q2	Q3	Q4	Q5
Flexibility only					
Mean $\widetilde{w}/w$	0.881	0.984	1.050	1.121	1.305
$\Pr(\hat{\mu} > 0)$	0.000	0.299	0.341	0.297	0.345
$\mathbb{E}[\hat{\mu} \mid \hat{\mu} > 0]$	–	0.149	0.209	0.219	0.237
Delegated task efficiency	–	1.000	0.998	0.991	0.988
Specialization only					
Mean $\widetilde{w}/w$	0.871	0.982	1.056	1.131	1.296
$\Pr(\hat{\mu} > 0)$	0.284	0.378	0.314	0.513	0.518
$\mathbb{E}[\hat{\mu} \mid \hat{\mu} > 0]$	0.192	0.199	0.197	0.204	0.209
Delegated task efficiency	1.351	1.351	1.351	1.351	1.351
With subcontracting					
Mean $\widetilde{w}/w$	0.860	0.972	1.052	1.124	1.325
$\Pr(\hat{\mu} > 0)$	0.423	0.375	0.314	0.427	0.498
$\mathbb{E}[\hat{\mu} \mid \hat{\mu} > 0]$	0.155	0.192	0.225	0.229	0.251
Delegated task efficiency	1.464	1.370	1.318	1.280	1.265

Appendix

Back Calibration

Same data, two models

	With subcontracting	⟶	Without subcontracting: Same parameters	⟶	Without subcontracting: Recalibrated
S.d. of sales growth	0.42		**0.28**		0.42
Comov. wage bill, rising sales	0.42		0.56		0.41
Comov. wage bill, falling sales	0.35		0.45		0.35

	Frictions		Shocks			Returns
	ψ_H	ψ_F	ρ	σ_η	σ_ε	ν
With subcontracting	1.61	2.61	0.983	0.044	0.078	0.908
Without subcontracting	**1.33**	**2.43**	0.973	0.055	**0.144**	0.872

⇒ Without subcontracting, sales far less volatile

⇒ Recalibration: larger transitory shocks, lower adjustment costs

Appendix

Back

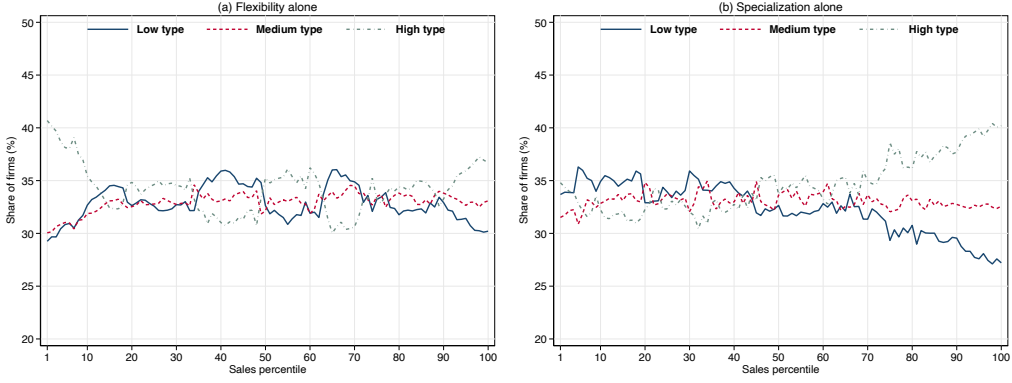
Calibration without subcontracting

Parameter		Value		Target	Data	Model	
		With	Without			With	Without
ρ	Persistence of z	0.983	0.973	Autocorr. log sales, 1 yr	0.952	0.948	0.926
σ_ε	Transitory shocks	0.078	0.144	S.d. sales growth, 1 yr	0.432	0.423	0.423
σ_η	Persistent innovations	0.044	0.055	S.d. sales growth, 3 yr	0.590	0.661	0.603
ψ_H	Hiring cost	1.61	1.33	Comov. wage bill, rising sales	0.415	0.421	0.414
ψ_F	Firing cost	2.61	2.43	Comov. wage bill, falling sales	0.353	0.350	0.350
ν	Returns to scale	0.908	0.872	Variable-cost share	0.917	0.880	0.854
μ_E	Entrants' mean draw	0.179	−0.023	Entrants' relative size	0.485	0.502	0.487
σ_E	Entrants' dispersion	0.100	0.123	Exit rate, entrants rel. to incumbents	1.521	1.514	1.521
δ_X	Exogenous exit	0.060	0.060	Firm exit rate	0.078	0.077	0.077
κ_F	Fixed cost	0.165	0.211	Exiters' relative size	0.776	0.776	0.773
κ_E	Entry cost	6.67	3.92	Mean employment	7.77	7.77	7.77

Appendix

Back

Type composition by size, flexibility only and specialization only

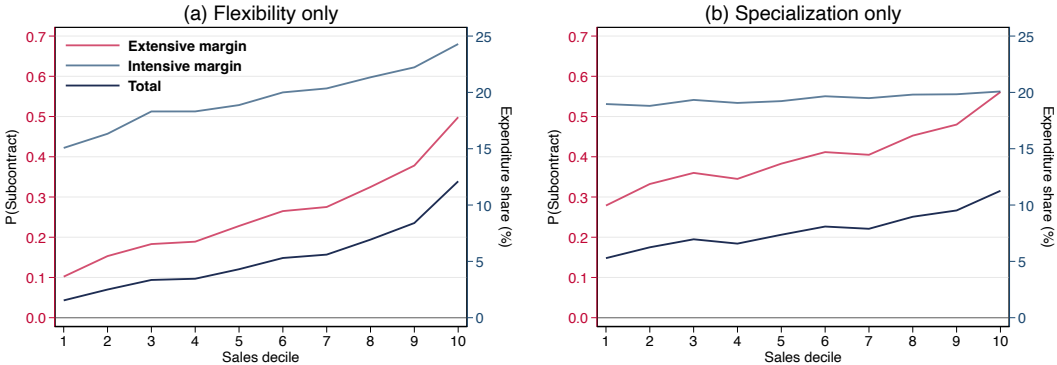


- Spec. only: high types concentrate among large firms, but less than in the baseline
- Flex. only: composition is flat; high types slightly more common at both ends, as they track shocks better

Appendix

Back

Subcontracting by size, flexibility only and specialization only



- Flexibility only: both margins rise with size, somewhat less than with subcontracting
- Specialization only: access rises with size, the expenditure share barely does

Appendix

Back

Responses to idiosyncratic and aggregate shocks: definitions

- Unexpected shock at $t = 0$: productivity multiplied by $A_t = \exp(\Delta_A \rho_A^t)$, with $\Delta_A = 0.01$

  Persistent: $\rho_A = 0.9$; transitory: $\rho_A = 0$

- Idiosyncratic: the shock hits one firm; w and ω stay at their steady-state values
- Aggregate: the shock hits all firms; w_t and ω_t follow the perfect-foresight transition

  Along the transition, entry responds to its value with a finite elasticity

- Firms active at $t = 0$, followed from then on (zero output after exit); y_{it}^0: their path without the shock

  Response at t : $r_t = \frac{1}{\Delta_A} \ln\left(\frac{\int_i y_{it}\, di}{\int_i y_{it}^0\, di}\right)$, on impact: r_0, cumulative: $\bar{r} = \sum_{t=0}^{T-1} \beta^t r_t$

- Labor and prices: same formula, with l_H, l_S, w_t or ω_t in place of output

Entry

Appendix

Back

Entry along the transition

- Steady state: free entry, $V_E = \kappa_E$, with a mass E^* of entrants
- Transition: free entry at every date makes entry, output and consumption swing
- Instead, potential entrants $\widetilde{E}$ with costs $\kappa \sim (\kappa/\bar{\kappa})^{\vartheta}$, entering if $\kappa \leq V_{E,t}$:

$$E_t = \widetilde{E}\left(\frac{V_{E,t}}{\bar{\kappa}}\right)^{\vartheta}$$

Elasticity ϑ: fixed entry if $\vartheta = 0$, free entry if $\vartheta \to \infty$; here $\vartheta = 2$

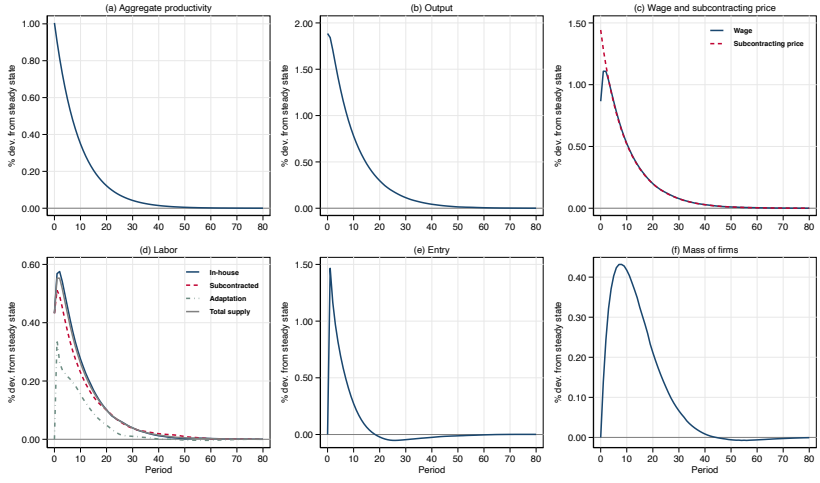
- Same steady state: the marginal entrant at E^* has cost κ_E

$$\widetilde{E}\left(\frac{\kappa_E}{\bar{\kappa}}\right)^{\vartheta} = E^* \quad \Longrightarrow \quad E_t = E^*\left(\frac{V_{E,t}}{\kappa_E}\right)^{\vartheta}$$

Appendix

Back

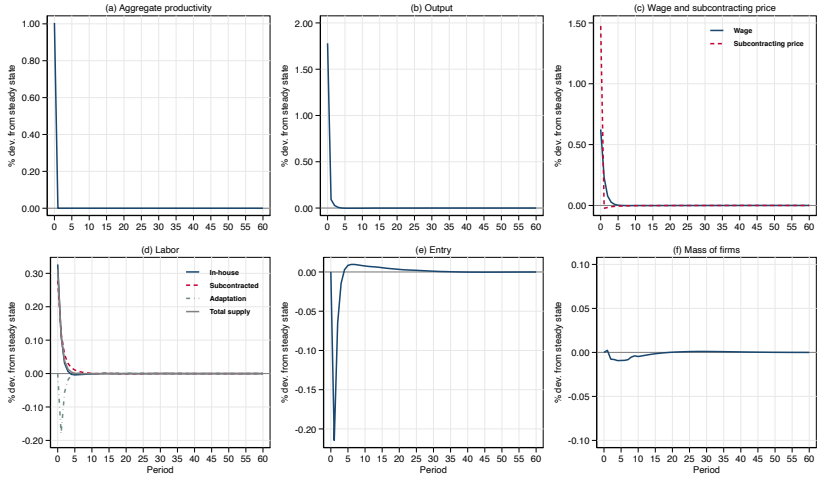
Transition after a 1% aggregate shock, persistent, $\rho_A = 0.9$



Appendix

Back

Transition after a 1% aggregate shock, transitory, $\rho_A = 0$



Appendix

Back

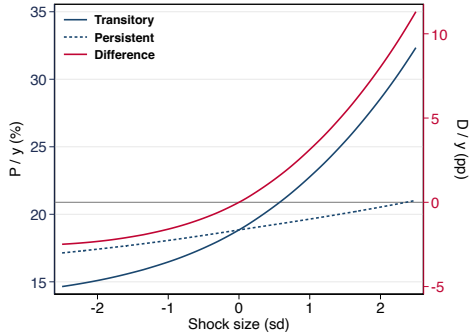
Responses to idiosyncratic and aggregate shocks

	Persistent			Transitory		
	Idio.	Agg.	Ratio	Idio.	Agg.	Ratio
Impact response, r_0						
With subcontracting	4.36	1.87	2.33	3.23	1.76	1.83
All firms with access	4.94	1.88	2.62	3.81	1.77	2.15
All firms without access	3.35	1.85	1.81	2.11	1.76	1.20
Without subcontracting	3.01	1.82	1.65	2.09	1.74	1.20
Cumulative response, $\bar{r}$						
With subcontracting	80.25	14.03	5.72	5.25	2.18	2.41
All firms with access	81.79	14.09	5.80	6.04	2.19	2.76
All firms without access	77.96	13.93	5.59	3.90	2.17	1.80
Without subcontracting	61.51	13.13	4.68	3.78	1.99	1.90

Appendix

Back

Value of access under transitory and persistent shocks



Notes: left axis, value of access relative to the firm's sales without the shock, averaged across firms; right axis, transitory minus persistent. Shock size in standard deviations of $\ln \varepsilon$.

- Persistent shocks: access worth 17–21% of yearly sales, almost flat
- Transitory shocks: from 15% of yearly sales for contractions to 29% for expansions

$\Rightarrow$ Flexibility: a demand for absorbing temporary movements

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